

Figure 7.45 Actual nonlinear system.

In Table 7.2 we list the response of the actual system for four different locations of the pole $z = a$, and an input $x(n) = \beta\delta(n)$, where $\beta = 15/16$, which has the binary representation 0.1111. Ideally, the response of the system should decay toward zero exponentially [i.e., $y(n) = a^n \rightarrow 0$ as $n \rightarrow \infty$]. In the actual system, however, the response $v(n)$ reaches a steady-state periodic output sequence with a period that depends on the value of the pole. When the pole is positive, the oscillations occur with a period $N_p = 1$, so that the output reaches a constant value of $\frac{1}{16}$ for $a = \frac{1}{2}$ and $\frac{1}{8}$ for $a = \frac{3}{4}$. On the other hand, when the pole is negative, the output sequence oscillates between positive and negative values ($\pm\frac{1}{16}$ for $a = -\frac{1}{2}$ and $\pm\frac{1}{8}$ for $a = -\frac{3}{4}$). Hence the period is $N_p = 2$.

These limit cycles occur as a result of the quantization effects in multiplications. When the input sequence $x(n)$ to the filter becomes zero, the output of the filter then, after a number of iterations, enters into the limit cycle. The output remains in the limit cycle until another input of sufficient size is applied that drives the system out of the limit cycle. Similarly, zero-input limit cycles occur from nonzero initial conditions with the input $x(n) = 0$. The amplitudes of the output during a limit cycle are confined to a range of values that is called the *dead band* of the filter.

TABLE 7.2 LIMIT CYCLES FOR LOWPASS SINGLE-POLE FILTER

n	$a = 0.1000$ $= \frac{1}{2}$	$a = 1.1000$ $= -\frac{1}{2}$	$a = 0.1100$ $= \frac{3}{4}$	$a = 1.1100$ $= -\frac{3}{4}$
0	0.1111 ($\frac{15}{16}$)	0.1111 ($\frac{15}{16}$)	0.1011 ($\frac{11}{16}$)	0.1011 ($\frac{11}{16}$)
1	0.1000 ($\frac{8}{16}$)	1.1000 ($-\frac{8}{16}$)	0.1000 ($\frac{8}{16}$)	1.1000 ($-\frac{8}{16}$)
2	0.0100 ($\frac{4}{16}$)	0.0100 ($\frac{4}{16}$)	0.0110 ($\frac{6}{16}$)	0.0110 ($\frac{6}{16}$)
3	0.0010 ($\frac{2}{16}$)	1.0010 ($-\frac{2}{16}$)	0.0101 ($\frac{5}{16}$)	1.0101 ($-\frac{5}{16}$)
4	0.0001 ($\frac{1}{16}$)	0.0001 ($\frac{1}{16}$)	0.0100 ($\frac{4}{16}$)	0.0100 ($\frac{4}{16}$)
5	0.0001 ($\frac{1}{16}$)	1.0001 ($-\frac{1}{16}$)	0.0011 ($\frac{3}{16}$)	1.0011 ($-\frac{3}{16}$)
6	0.0001 ($\frac{1}{16}$)	0.0001 ($\frac{1}{16}$)	0.0010 ($\frac{2}{16}$)	0.0010 ($\frac{2}{16}$)
7	0.0001 ($\frac{1}{16}$)	1.0001 ($-\frac{1}{16}$)	0.0010 ($\frac{2}{16}$)	1.0010 ($-\frac{2}{16}$)
8	0.0001 ($\frac{1}{16}$)	0.0001 ($\frac{1}{16}$)	0.0010 ($\frac{2}{16}$)	0.0010 ($\frac{2}{16}$)

It is interesting to note that when the response of the single-pole filter is in the limit cycle, the actual nonlinear system operates as an equivalent linear system with a pole at $z = 1$ when the pole is positive and $z = -1$ when the pole is negative. That is,

$$Q_r[av(n-1)] = \begin{cases} v(n-1), & a > 0 \\ -v(n-1), & a < 0 \end{cases} \quad (7.7.3)$$

Since the quantized product $av(n-1)$ is obtained by rounding, it follows that the quantization error is bounded as

$$|Q_r[av(n-1)] - av(n-1)| \leq \frac{1}{2} \cdot 2^{-b} \quad (7.7.4)$$

where b is the number of bits (exclusive of sign) used in the representation of the pole a and $v(n)$. Consequently, (7.7.4) and (7.7.3) leads to

$$|v(n-1)| - |av(n-1)| \leq \frac{1}{2} \cdot 2^{-b}$$

and hence

$$|v(n-1)| \leq \frac{\frac{1}{2} \cdot 2^{-b}}{1 - |a|} \quad (7.7.5)$$

The expression in (7.7.5) defines the dead band for a single-pole filter. For example, when $b = 4$ and $|a| = \frac{1}{2}$, we have a dead band with a range of amplitudes $(-\frac{1}{16}, \frac{1}{16})$. When $b = 4$ and $|a| = \frac{3}{4}$, the dead band increases to $(-\frac{1}{8}, \frac{1}{8})$.

The limit-cycle behavior in a two-pole filter is much more complex and a larger variety of oscillations can occur. In this case the ideal two-pole system is described by the linear difference equation,

$$y(n) = a_1 y(n-1) + a_2 y(n-2) + x(n) \quad (7.7.6)$$

whereas the actual system is described by the nonlinear difference equation

$$v(n) = Q_r[a_1 v(n-1)] + Q_r[a_2 v(n-2)] + x(n) \quad (7.7.7)$$

When the filter coefficients satisfy the condition $a_1^2 < -4a_2$, the poles of the system occur at

$$z = re^{\pm j\theta}$$

where $a_2 = -r^2$ and $a_1 = 2r \cos \theta$. As in the case of the single-pole filter, when the system is in a zero-input or zero-state limit cycle,

$$Q_r[a_2 v(n-2)] = -v(n-2) \quad (7.7.8)$$

In other words, the system behaves as an oscillator with complex-conjugate poles on the unit circle (i.e., $a_2 = -r^2 = -1$). Rounding the product $av(n-2)$ implies that

$$|Q_r[a_2 v(n-2)] - a_2 v(n-2)| \leq \frac{1}{2} \cdot 2^{-b} \quad (7.7.9)$$

Upon substitution of (7.7.8) into (7.7.9), we obtain the result

$$|v(n-2)| - |a_2 v(n-2)| \leq \frac{1}{2} \cdot 2^{-b}$$

or equivalently,

$$|v(n-2)| \leq \frac{\frac{1}{2} \cdot 2^{-b}}{1 - |a_2|} \quad (7.7.10)$$

The expression in (7.7.10) defines the dead band of the two-pole filter with complex-conjugate poles. We observe that the dead-band limits depend only on $|a_2|$. The parameter $a_1 = 2r \cos \theta$ determines the frequency of oscillation.

Another possible limit-cycle mode with zero input, which occurs as a result of rounding the multiplications, corresponds to an equivalent second-order system with poles at $z = \pm 1$. In this case it was shown by Jackson (1969) that the two-pole filter exhibits oscillations with an amplitude that falls in the dead band bounded by $2^{-b}/(1 - |a_1| - a_2)$.

It is interesting to note that these limit cycles result from rounding the product of the filter coefficients with the previous outputs, $v(n-1)$ and $v(n-2)$. Instead of rounding, we may choose to truncate the products to b bits. With truncation, we can eliminate many, although not all, of the limit cycles as shown by Claasen et al. (1973). However, recall that truncation results in a biased error unless the sign-magnitude representation is used, in which case the truncation error is symmetric about zero. In general, this bias is undesirable in digital filter implementation.

In a parallel realization of a high-order IIR system, each second-order filter section exhibits its own limit-cycle behavior, with no interaction among the second-order filter sections. Consequently, the output is the sum of the zero-input limit cycles from the individual sections. In the case of a cascade realization for a high-order IIR system, the limit cycles are much more difficult to analyze. In particular, when the first filter section exhibits a zero-input limit cycle, the output limit cycle is filtered by the succeeding sections. If the frequency of the limit cycle falls near a resonance frequency in a succeeding filter section, the amplitude of the sequence is enhanced by the resonance characteristic. In general, we must be careful to avoid such situations.

In addition to limit cycles caused by rounding the result of multiplications, there are limit cycles caused by overflows in addition. An overflow in addition of two or more binary numbers occurs when the sum exceeds the word size available in the digital implementation of the system. For example, let us consider the second-order filter section illustrated in Fig. 7.46, in which the addition is performed in two's-complement arithmetic. Thus we can write the output $y(n)$ as

$$y(n) = g[a_1 y(n-1) + a_2 y(n-2) + x(n)] \quad (7.7.11)$$

where the function $g[\cdot]$ represents the two's-complement addition. It is easily verified that the function $g(v)$ versus v is described by the graph in Fig. 7.47.

Recall that the range of values of the parameters (a_1, a_2) for a stable filter is given by the stability triangle in Fig. 3.15. However, these conditions are no longer sufficient to prevent overflow oscillation with two's-complement arithmetic.

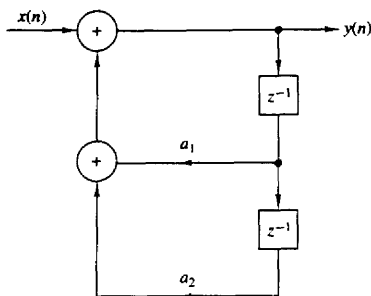


Figure 7.46 Two-pole filter realization.

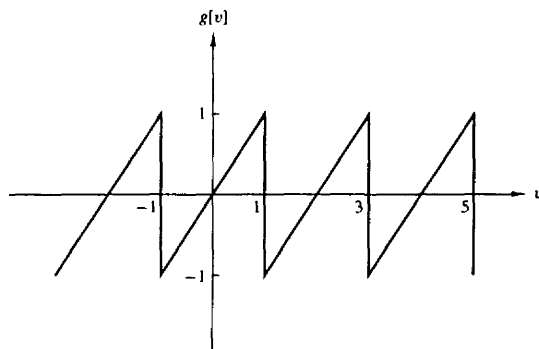


Figure 7.47 Characteristic functional relationship for two's complement addition of two or more numbers.

In fact, it can easily be shown that a necessary and sufficient condition for ensuring that no zero-input overflow limit cycles occur is

$$|a_1| + |a_2| < 1 \quad (7.7.12)$$

which is extremely restrictive and hence an unreasonable constraint to impose on any second-order section.

An effective remedy for curing the problem of overflow oscillations is to modify the adder characteristic, as illustrated in Fig. 7.48, so that it performs saturation arithmetic. Thus when an overflow (or underflow) is sensed, the output of the adder will be the full-scale value of ± 1 . The distortion caused by this nonlinearity in the adder is usually small provided that saturation occurs infrequently. The use of such a nonlinearity does not preclude the need for scaling of the signals and the system parameters, as described in the following section.

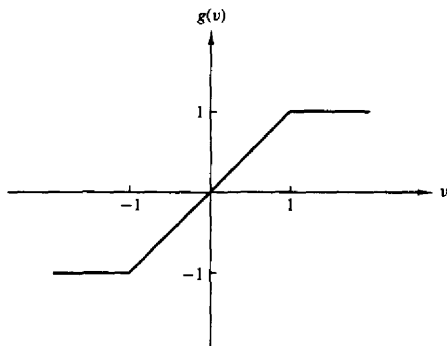


Figure 7.48 Characteristic functional relationship for addition with clipping at ± 1 .

7.7.2 Scaling to Prevent Overflow

Saturation arithmetic as just described eliminates limit cycles due to overflow, on the one hand, but on the other hand, it causes undesirable signal distortion due to the nonlinearity of the clipper. In order to limit the amount of nonlinear distortion, it is important to scale the input signal and the unit sample response, between the input and any internal summing node in the system, such that overflow becomes a rare event.

For fixed-point arithmetic, let us first consider the extreme condition that overflow is not permitted at any node of the system. Let $y_k(n)$ denote the response of the system at the k th node when the input sequence is $x(n)$ and the unit sample response between the node and the input is $h_k(n)$. Then

$$|y_k(n)| = \left| \sum_{m=-\infty}^{\infty} h_k(m)x(n-m) \right| \leq \sum_{m=-\infty}^{\infty} |h_k(m)||x(n-m)|$$

Suppose that $x(n)$ is upper bounded by A_x . Then

$$|y_k(n)| \leq A_x \sum_{m=-\infty}^{\infty} |h_k(m)| \quad \text{for all } n \quad (7.7.13)$$

Now, if the dynamic range of the computer is limited to $(-1, 1)$, the condition

$$|y_k(n)| < 1$$

can be satisfied by requiring that the input $x(n)$ be scaled such that

$$A_x < \frac{1}{\sum_{m=-\infty}^{\infty} |h_k(m)|} \quad (7.7.14)$$

for all possible nodes in the system. The condition in (7.7.14) is both necessary and sufficient to prevent overflow.

The condition in (7.7.14) is overly conservative, however, to the point where the input signal may be scaled too much. In such a case, much of the precision used to represent $x(n)$ is lost. This is especially true for narrowband sequences, such as sinusoids, where the scaling implied by (7.7.14) is extremely severe. For narrowband signals we can use the frequency response characteristics of the system in determining the appropriate scaling. Since $|H(\omega)|$ represents the gain of the system at frequency ω , a less severe and reasonably adequate scaling is to require that

$$A_x < \frac{1}{\max_{0 \leq \omega \leq \pi} |H_k(\omega)|} \quad (7.7.15)$$

where $H_k(\omega)$ is the Fourier transform of $\{h_k(n)\}$.

In the case of an FIR filter, the condition in (7.7.14) reduces to

$$A_x < \frac{1}{\sum_{m=0}^{M-1} |h_k(m)|} \quad (7.7.16)$$

which is now a sum over the M nonzero terms of the filter unit sample response.

Another approach to scaling is to scale the input so that

$$\sum_{n=-\infty}^{\infty} |y_k(n)|^2 \leq C^2 \sum_{n=-\infty}^{\infty} |x(n)|^2 = C^2 E_x \quad (7.7.17)$$

From Parseval's theorem we have

$$\begin{aligned} \sum_{n=-\infty}^{\infty} |y_k(n)|^2 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(\omega)X(\omega)|^2 d\omega \\ &\leq E_x \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(\omega)|^2 d\omega \end{aligned} \quad (7.7.18)$$

By combining (7.7.17) with (7.7.18), we obtain

$$C^2 \leq \frac{1}{\sum_{n=-\infty}^{\infty} |h_k(n)|^2} = \frac{1}{(1/2\pi) \int_{-\pi}^{\pi} |H(\omega)|^2 d\omega} \quad (7.7.19)$$

If we compare the different scaling factors given above, we find that

$$\left[\sum_{n=-\infty}^{\infty} |h_k(n)|^2 \right]^{1/2} \leq \max_{\omega} |H_k(\omega)| \leq \sum_{n=-\infty}^{\infty} |h_k(n)| \quad (7.7.20)$$

Clearly, (7.7.14) is the most pessimistic constraint.

In the following section we observe the ramifications of this scaling on the output signal-to-noise (power) ratio (SNR) from a first-order and a second-order filter section.

7.7.3 Statistical Characterization of Quantization Effects in Fixed-Point Realizations of Digital Filters

It is apparent from our treatment in the previous section that an analysis of quantization errors in digital filtering, based on deterministic models of quantization effects, is not a very fruitful approach. The basic problem is that the nonlinear effects in quantizing the products of two numbers and in clipping the sum of two numbers to prevent overflow are not easily modeled in large systems that contain many multipliers and many summing nodes.

To obtain more general results on the quantization effects in digital filters, we shall model the quantization errors in multiplication as an additive noise sequence $e(n)$, just as we did in characterizing the quantization errors in A/D conversion of an analog signal. For addition, we consider the effect of scaling the input signal to prevent overflow.

Let us begin our treatment with the characterization of the round-off noise in a single-pole filter which is implemented in fixed-point arithmetic and is described by the nonlinear difference equation

$$v(n) = Q_r[av(n-1)] + x(n) \quad (7.7.21)$$

The effect of rounding the product $av(n-1)$ is modeled as a noise sequence $e(n)$ added to the actual product $av(n-1)$, that is,

$$Q_r[av(n-1)] = av(n-1) + e(n) \quad (7.7.22)$$

With this model for the quantization error, the system under consideration is described by the *linear difference equation*

$$v(n) = av(n-1) + x(n) + e(n) \quad (7.7.23)$$

The corresponding system is illustrated in block diagram form in Fig. 7.49.

It is apparent from (7.7.23) that the output sequence $v(n)$ of the filter can be separated into two components. One is the response of the system to the input sequence $x(n)$. The second is the response of the system to the additive quantization noise $e(n)$. In fact, we can express the output sequence $v(n)$ as a sum of these two components, that is,

$$v(n) = y(n) + q(n) \quad (7.7.24)$$

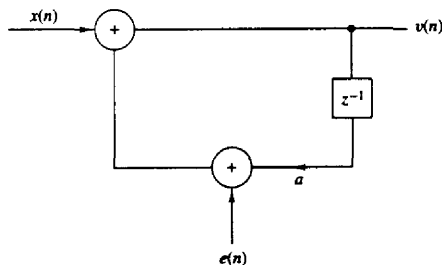


Figure 7.49 Additive noise model for the quantization error in a single-pole filter.

where $y(n)$ represents the response of the system to $x(n)$, and $q(n)$ represents the response of the system to the quantization error $e(n)$. Upon substitution from (7.7.24) for $v(n)$ into (7.7.23), we obtain

$$y(n) + q(n) = ay(n-1) + aq(n-1) + x(n) + e(n) \quad (7.7.25)$$

To simplify the analysis, we make the following assumptions about the error sequence $e(n)$.

1. For any n , the error sequence $\{e(n)\}$ is uniformly distributed over the range $(-\frac{1}{2}, 2^{-b}, \frac{1}{2} \cdot 2^{-b})$. This implies that the mean value of $e(n)$ is zero and its variance is

$$\sigma_e^2 = \frac{2^{-2b}}{12} \quad (7.7.26)$$

2. The error $\{e(n)\}$ is a stationary white noise sequence. In other words, the error $e(n)$ and the error $e(m)$ are uncorrelated for $n \neq m$.
3. The error sequence $\{e(n)\}$ is uncorrelated with the signal sequence $\{x(n)\}$.

The last assumption allows us to separate the difference equation in (7.7.25) into two uncoupled difference equations, namely,

$$y(n) = ay(n-1) + x(n) \quad (7.7.27)$$

$$q(n) = aq(n-1) + e(n) \quad (7.7.28)$$

The difference equation in (7.7.27) represents the input-output relation for the desired system and the difference equation in (7.7.28) represents the relation for the quantization error at the output of the system.

To complete the analysis, we make use of two important relationships developed in Appendix A. The first is the relationship for the mean value of the output $q(n)$ of a linear shift-invariant filter with impulse response $h(n)$ when excited by a random sequence $e(n)$ having a mean value m_e . The result is

$$m_q = m_e \sum_{n=-\infty}^{\infty} h(n) \quad (7.7.29)$$

or, equivalently,

$$m_q = m_e H(0) \quad (7.7.30)$$

where $H(0)$ is the value of the frequency response $H(\omega)$ of the filter evaluated at $\omega = 0$.

The second important relationship is the expression for the autocorrelation sequence of the output $q(n)$ of the filter with impulse response $h(n)$ when the input random sequence $e(n)$ has an autocorrelation $\gamma_{ee}(n)$. This result is

$$\gamma_{qq}(n) = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} h(k)h(l)\gamma_{ee}(k-l+n) \quad (7.7.31)$$

In the important special case where the random sequence is white (spectrally flat), the autocorrelation $\gamma_{ee}(n)$ is a unit sample sequence scaled by the variance σ_e^2 , that is,

$$\gamma_{ee}(n) = \sigma_e^2 \delta(n) \quad (7.7.32)$$

Upon substituting (7.7.32) into (7.7.31), we obtain the desired result for the autocorrelation sequence at the output of a filter excited by white noise, namely,

$$\gamma_{qq}(n) = \sigma_e^2 \sum_{k=-\infty}^{\infty} h(k)h(k+n) \quad (7.7.33)$$

The variance σ_q^2 of the output noise is simply obtained by evaluating $\gamma_{qq}(n)$ at $n = 0$. Thus

$$\sigma_q^2 = \sigma_e^2 \sum_{k=-\infty}^{\infty} h^2(k) \quad (7.7.34)$$

and with the aid of Parseval's theorem, we have the alternative expression

$$\sigma_q^2 = \frac{\sigma_e^2}{2\pi} \int_{-\pi}^{\pi} |H(\omega)|^2 d\omega \quad (7.7.35)$$

In the case of the single-pole filter under consideration, the unit sample response is

$$h(n) = a^n u(n) \quad (7.7.36)$$

Since the quantization error due to rounding has zero mean, the mean value of the error at the output of the filter is $m_q = 0$. The variance of the error at the output of the filter is

$$\begin{aligned} \sigma_q^2 &= \sigma_e^2 \sum_{k=0}^{\infty} a^{2k} \\ &= \frac{\sigma_e^2}{1-a^2} \end{aligned} \quad (7.7.37)$$

We observe that the noise power σ_q^2 at the output of the filter is enhanced relative to the input noise power σ_e^2 by the factor $1/(1-a^2)$. This factor increases as the pole is moved closer to the unit circle.

To obtain a clearer picture of the effect of the quantization error, we should also consider the effect of scaling the input. Let us assume that the input sequence $\{x(n)\}$ is a white noise sequence (wideband signal), whose amplitude has been scaled according to (7.7.14) to prevent overflows in addition. Then

$$A_x < 1 - |a|$$

If we assume that $x(n)$ is uniformly distributed in the range $(-A_x, A_x)$, then, according to (7.7.31) and (7.7.34), the signal power at the output of the filter

is

$$\begin{aligned}\sigma_y^2 &= \sigma_x^2 \sum_{k=0}^{\infty} a^{2k} \\ &= \frac{\sigma_x^2}{1-a^2}\end{aligned}\quad (7.7.38)$$

where $\sigma_x^2 = (1-|a|)^2/3$ is the variance of the input signal. The ratio of the signal power σ_y^2 to the quantization error power σ_q^2 , which is called the signal-to-noise ratio (SNR), is simply

$$\begin{aligned}\frac{\sigma_y^2}{\sigma_q^2} &= \frac{\sigma_x^2}{\sigma_e^2} \\ &= (1-|a|)^2 \cdot 2^{2(b+1)}\end{aligned}\quad (7.7.39)$$

This expression for the output SNR clearly illustrates the severe penalty paid as a consequence of the scaling of the input, especially when the pole is near the unit circle. By comparison, if the input is not scaled and the adder has a sufficient number of bits to avoid overflow, then the signal amplitude may be confined to the range $(-1, 1)$. In this case, $\sigma_x^2 = \frac{1}{3}$, which is independent of the pole position. Then

$$\frac{\sigma_y^2}{\sigma_q^2} = 2^{2(b+1)} \quad (7.7.40)$$

The difference between the SNRs in (7.7.40) and (7.7.39) clearly demonstrates the need to use more bits in addition than in multiplication. The number of additional bits depends on the position of the pole and should be increased as the pole is moved closer to the unit circle.

Next, let us consider a two-pole filter with infinite precision which is described by the linear difference equation

$$y(n) = a_1 y(n-1) + a_2 y(n-2) + x(n) \quad (7.7.41)$$

where $a_1 = 2r \cos \theta$ and $a_2 = -r^2$. When the two products are rounded, we have a system which is described by the nonlinear difference equation

$$v(n) = Q_r[a_1 v(n-1)] + Q_r[a_2 v(n-2)] + x(n) \quad (7.7.42)$$

This system is illustrated in block diagram form in Fig. 7.50.

Now there are two multiplications, and hence two quantization errors are produced for each output. Consequently, we should introduce two noise sequences $e_1(n)$ and $e_2(n)$, which correspond to the quantizer outputs

$$\begin{aligned}Q_r[a_1 v(n-1)] &= a_1 v(n-1) + e_1(n) \\ Q_r[a_2 v(n-2)] &= a_2 v(n-2) + e_2(n)\end{aligned}\quad (7.7.43)$$

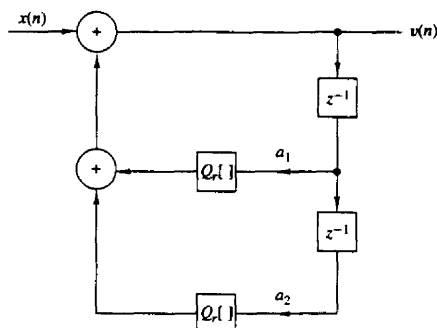


Figure 7.50 Two-pole digital filter with rounding quantizers.

A block diagram for the corresponding model is shown in Fig. 7.51. Note that the error sequences $e_1(n)$ and $e_2(n)$ can be moved directly to the input of the filter.

As in the case of the first-order filter, the output of the second-order filter can be separated into two components, the desired signal component and the quantization error component. The former is described by the difference equation

$$y(n) = a_1 y(n-1) + a_2 y(n-2) + x(n) \quad (7.7.44)$$

while the latter satisfies the difference equation

$$q(n) = a_1 q(n-1) + a_2 q(n-2) + e_1(n) + e_2(n) \quad (7.7.45)$$

It is reasonable to assume that the two sequences $e_1(n)$ and $e_2(n)$ are uncorrelated.

Now the second-order filter has a unit sample response

$$h(n) = \frac{r^n}{\sin \theta} \sin(n+1)\theta u(n) \quad (7.7.46)$$

Hence

$$\sum_{n=0}^{\infty} h^2(n) = \frac{1+r^2}{1-r^2} \frac{1}{r^4 + 1 - 2r^2 \cos 2\theta} \quad (7.7.47)$$

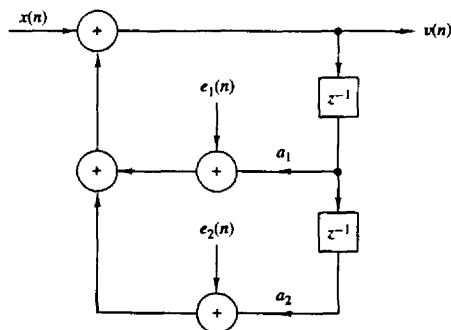


Figure 7.51 Additive noise model for the quantization errors in a two-pole filter realization.

By applying (7.7.34), we obtain the variance of the quantization errors at the output of the filter in the form

$$\sigma_q^2 = \sigma_e^2 \left(\frac{1+r^2}{1-r^2} \frac{1}{r^4 + 1 - 2r^2 \cos 2\theta} \right) \quad (7.7.48)$$

In the case of the signal component, if we scale the input as in (7.7.14) to avoid overflow, the power in the output signal is

$$\sigma_y^2 = \sigma_x^2 \sum_{n=0}^{\infty} h^2(n) \quad (7.7.49)$$

where the power in the input signal $x(n)$ is given by the variance

$$\sigma_x^2 = \frac{1}{3 \left[\sum_{n=0}^{\infty} |h(n)| \right]^2} \quad (7.7.50)$$

Consequently, the SNR at the output of the two-pole filter is

$$\frac{\sigma_y^2}{\sigma_q^2} = \frac{\sigma_x^2}{\sigma_e^2} = \frac{2^{2(b+1)}}{\left[\sum_{n=0}^{\infty} |h(n)| \right]^2} \quad (7.7.51)$$

Although it is difficult to determine the exact value of the denominator term in (7.7.51), it is easy to obtain an upper and a lower bound. In particular, $|h(n)|$ is upper bounded as

$$|h(n)| \leq \frac{1}{\sin \theta} r^n \quad n \geq 0 \quad (7.7.52)$$

so that

$$\sum_{n=0}^{\infty} |h(n)| \leq \frac{1}{\sin \theta} \sum_{n=0}^{\infty} r^n = \frac{1}{(1-r) \sin \theta} \quad (7.7.53)$$

The lower bound may be obtained by noting that

$$|H(\omega)| = \left| \sum_{n=0}^{\infty} h(n) e^{-j\omega n} \right| \leq \sum_{n=0}^{\infty} |h(n)|$$

But

$$H(\omega) = \frac{1}{(1 - re^{j\theta} e^{-j\omega})(1 - re^{-j\theta} e^{-j\omega})}$$

At $\omega = \theta$, which is the resonant frequency of the filter, we obtain the largest value of $|H(\omega)|$. Hence

$$\sum_{n=0}^{\infty} |h(n)| \geq |H(\theta)| = \frac{1}{(1-r)\sqrt{1+r^2-2r\cos 2\theta}} \quad (7.7.54)$$

Therefore, the SNR is bounded from above and below according to the relation

$$2^{2(b+1)}(1-r)^2 \sin^2 \theta \leq \frac{\sigma_y^2}{\sigma_q^2} \leq 2^{2(b+1)}(1-r)^2(1+r^2-2r \cos 2\theta) \quad (7.7.55)$$

For example, when $\theta = \pi/2$, the expression in (7.7.55) reduces to

$$2^{2(b+1)}(1-r)^2 \leq \frac{\sigma_y^2}{\sigma_q^2} \leq 2^{2(b+1)}(1-r)^2(1+r)^2 \quad (7.7.56)$$

The dominant term in this bound is $(1-r)^2$ which acts to reduce the SNR dramatically as the poles move toward the unit circle. Hence the effect of scaling in the second-order filter is more severe than in the single-pole filter. Note that if $d = 1-r$ is the distance of the pole from the unit circle, the SNR in (7.7.56) is reduced by d^2 , whereas in the single-pole filter the reduction is proportional to d . These results serve to reinforce the earlier statement regarding the use of more bits in addition than in multiplication as a mechanism for avoiding the severe penalty due to scaling.

The analysis of the quantization effects in a second-order filter can be applied directly to higher-order filters based on a parallel realization. In this case each second-order filter section is independent of all the other sections, and therefore the total quantization noise power at the output of the parallel bank is simply the linear sum of the quantization noise powers of each of the individual sections. On the other hand, the cascade realization is more difficult to analyze. For the cascade interconnection, the noise generated in any second-order filter section is filtered by the succeeding sections. As a consequence, there is the issue of how to pair together real-valued poles to form second-order sections and how to arrange the resulting second-order filters to minimize the total noise power at the output of the high-order filter. This general topic was investigated by Jackson (1970a, b), who showed that poles close to the unit circle should be paired with nearby zeros to reduce the gain of each second-order section. In ordering the second-order sections in cascade, a reasonable strategy is to place the sections in the order of decreasing maximum frequency gain. In this case the noise power generated in the early high-gain section is not boosted significantly by the latter sections.

The following example illustrates the point that proper ordering of sections in a cascade realization is important in controlling the round-off noise at the output of the overall filter.

Example 7.7.1

Determine the variance of the round-off noise at the output of the two cascade realizations of the filter with system function

$$H(z) = H_1(z)H_2(z)$$

where

$$H_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

$$H_2(z) = \frac{1}{1 - \frac{1}{4}z^{-1}}$$

Solution Let $h(n)$, $h_1(n)$, and $h_2(n)$ represent the unit sample responses corresponding to the system functions $H(z)$, $H_1(z)$, and $H_2(z)$, respectively. It follows that

$$h_1(n) = \left(\frac{1}{2}\right)^n u(n) \quad h_2(n) = \left(\frac{1}{4}\right)^n u(n)$$

$$h(n) = [2\left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n] u(n)$$

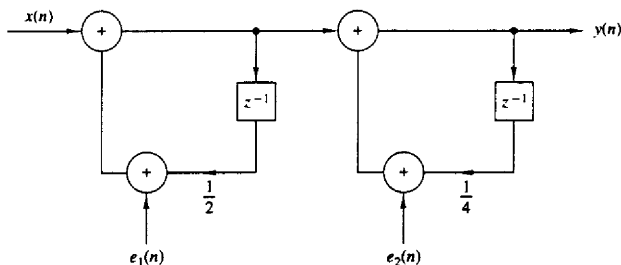
The two cascade realizations are shown in Fig. 7.52.

In the first cascade realization, the variance of the output is

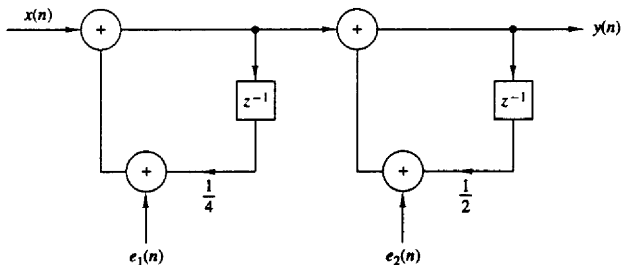
$$\sigma_{q1}^2 = \sigma_r^2 \left[\sum_{n=0}^{\infty} h^2(n) + \sum_{n=0}^{\infty} h_2^2(n) \right]$$

In the second cascade realization, the variance of the output noise is

$$\sigma_{q2}^2 = \sigma_r^2 \left[\sum_{n=0}^{\infty} h^2(n) + \sum_{n=0}^{\infty} h_1^2(n) \right]$$



(a) Cascade realization I



(b) Cascade realization II

Figure 7.52 Two cascade realizations in Example 7.8.1: (a) cascade realization I; (b) cascade realization II.

Now

$$\sum_{n=0}^{\infty} h_1^2(n) = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}$$

$$\sum_{n=0}^{\infty} h_2^2(n) = \frac{1}{1 - \frac{1}{16}} = \frac{16}{15}$$

$$\sum_{n=0}^{\infty} h^2(n) = \frac{4}{1 - \frac{1}{4}} - \frac{4}{1 - \frac{1}{8}} + \frac{1}{1 - \frac{1}{16}} = 1.83$$

Therefore,

$$\sigma_{q1}^2 = 2.90\sigma_e^2$$

$$\sigma_{q2}^2 = 3.16\sigma_e^2$$

and the ratio of noise variances is

$$\frac{\sigma_{q2}^2}{\sigma_{q1}^2} = 1.09$$

Consequently, the noise power in the second cascade realization is 9% larger than the first realization.

7.8 SUMMARY AND REFERENCES

From the treatment in this chapter we have seen that there are various realizations of discrete-time systems. FIR systems can be realized in a direct form, a cascade form, a frequency sampling form, and a lattice form. IIR systems can also be realized in a direct form, a cascade form, a lattice or a lattice-ladder form, and in a parallel form.

For any given system described by a linear constant-coefficient difference equation, these realizations are equivalent in that they represent the same system and produce the same output for any given input, provided that the internal computations are performed with infinite precision. However, the various structures are not equivalent when they are realized with finite-precision arithmetic.

The state-space formulation provides an internal description of a system and, as a consequence, we obtained additional system realizations, called *state-space realizations*. These realizations represent additional possible structures that provide good alternative candidate realizations for the system.

Three important factors are presented for choosing among the various FIR and IIR system realizations. These factors are computational complexity, memory requirements, and finite-word-length effects. Depending on either the time-domain or the frequency-domain characteristics of a system, some structures may require less computation and/or less memory than others. Hence our selection must consider these two important factors.

Much research has been done over the past two decades on state-space representation and realization of systems. For reference, we cite the books by Chen

(1970), DeRusso et al. (1965), Zadeh and Desoer (1963), and Gupta (1966). The use of state-space filter structures in the realization of IIR systems has been proposed by Mullis and Roberts (1976a,b), and further developed by Hwang (1977), Jackson et al. (1979), Jackson (1979), Mills et al. (1981), and Bomar (1985).

In deriving the transposed structures in Section 7.3, we introduced several concepts and operations on signal flow graphs. Signal flow graphs are treated in depth in the books by Mason and Zimmerman (1960) and Chow and Cassignol (1962).

Another important structure for IIR systems, a *wave digital filter*, has been investigated by Fettweis (1971) and further developed by Sedlmeyer and Fettweis (1973). A treatment of this filter structure can also be found in the book by Antoniou (1979).

Finite-word-length effects are an important factor in the implementation of digital signal processing systems. In this chapter we described the effects of a finite word length in digital filtering. In particular, we considered the following problems dealing with finite-word length effects:

1. Parameter quantization in digital filters
2. Round-off noise in multiplication
3. Overflow in addition
4. Limit cycles

These four effects are internal to the filter and influence the method by which the system will be implemented. In particular, we demonstrated that high-order systems, especially IIR systems, should be realized by using second-order sections as building blocks. We advocated the use of the direct form II realization, either the conventional or the transposed form.

Effects of round-off errors in fixed-point implementations of FIR and IIR filter structures have been investigated by many researchers. We cite the papers by Gold and Rader (1966), Rader and Gold (1967b), Jackson (1970a,b), Liu (1971), Chan and Rabiner (1973a,b,c), and Oppenheim and Weinstein (1972).

As an alternative to the use of direct form II second-order filters as building blocks for high-order filters, we can use second-order state-variable forms. Such state-variable forms can be optimized with respect to the state transition matrix to minimize round-off errors. The optimization leads to minimum-round-off-noise second-order state-variable filters that are highly robust for implementing both narrowband and wideband filters.

For a treatment of minimum-round-off-noise second-order state-space realizations, the reader can refer to the papers of Mullis and Roberts (1976a,b), Hwang (1977), Jackson et al. (1979), Mills et al. (1981), Bomar (1985), and the book by Roberts and Mullis (1987).

Limit-cycle oscillations occur in IIR filters as a result of quantization effects in fixed-point multiplication and rounding. Investigation of limit cycles in digital filtering and their characteristic behavior is treated in the papers by Parker and

Hess (1971), Brubaker and Gowdy (1972), Sandberg and Kaiser (1972), and Jackson (1969, 1979). The latter paper deals with limit cycles in state-space structures. Methods have also been devised to eliminate limit cycles caused by round-off errors. For example, the papers by Barnes and Fam (1977), Fam and Barnes (1979), Chang (1981), Butterweck et al. (1984), and Auer (1987) discuss this problem. Overflow oscillations have been treated in the paper by Ebert et al. (1969).

The effects of parameter quantization has been treated in a number of papers. We cite for reference the work of Rader and Gold (1967b), Knowles and Olcayto (1968), Avenhaus and Schuessler (1970), Herrmann and Schuessler (1970b), Chan and Rabiner (1973c), and Jackson (1976).

Finally, we mention that the lattice and lattice-ladder filter structures are known to be robust in fixed-point implementations. For a treatment of these types of filters, the reader is referred to the papers of Gray and Markel (1973), Makhoul (1978), and Morf et al. (1977) and to the book by Markel and Gray (1976).

PROBLEMS

7.1 Determine a direct form realization for the following linear phase filters.

(a) $h(n) = \{1, 2, 3, 4, 3, 2, 1\}$

↑

(b) $h(n) = \{1, 2, 3, 3, 2, 1\}$

↑

7.2 Consider an FIR filter with system function

$$H(z) = 1 + 2.88z^{-1} + 3.4048z^{-2} + 1.74z^{-3} + 0.4z^{-4}$$

Sketch the direct form and lattice realizations of the filter and determine in detail the corresponding input-output equations. Is the system minimum phase?

7.3 Determine the system function and the impulse response of the system shown in Fig. P7.3.

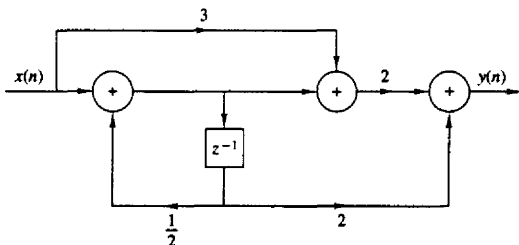


Figure P7.3

7.4 Determine the system function and the impulse response of the system shown in Fig. P7.4.

7.5 Determine the transposed structure of the systems in Fig. P7.4 and verify that both the original and the transposed system have the same system function.

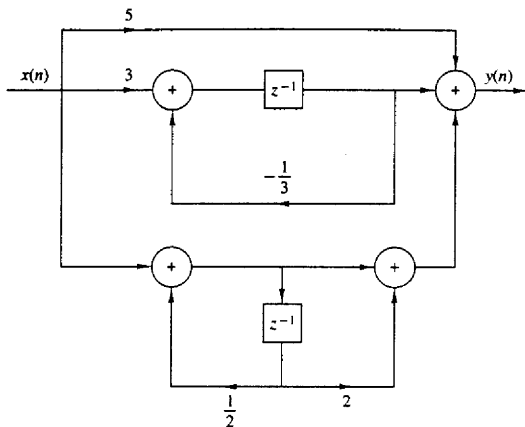


Figure P7.4

- 7.6 Determine a_1 , a_2 , and c_1 and c_0 in terms of b_1 and b_2 so that the two systems in Fig. P7.6 are equivalent.
- 7.7 Consider the filter shown in Fig. P7.7.
- Determine its system function.
 - Sketch the pole-zero plot and check for stability if
 - $b_0 = b_2 = 1$, $b_1 = 2$, $a_1 = 1.5$, $a_2 = -0.9$
 - $b_0 = b_2 = 1$, $b_1 = 2$, $a_1 = 1$, $a_2 = -2$
 - Determine the response to $x(n) = \cos(\pi n/3)$ if $b_0 = 1$, $b_1 = b_2 = 0$, $a_1 = 1$, and $a_2 = -0.99$.
- 7.8 Consider an LTI system, initially at rest, described by the difference equation

$$y(n) = \frac{1}{4}y(n-2) + x(n)$$

- Determine the impulse response, $h(n)$, of the system.
 - What is the response of the system to the input signal

$$x(n) = \left[\left(\frac{1}{2}\right)^n + \left(-\frac{1}{2}\right)^n\right]u(n)$$
 - Determine the direct form II, parallel-form, and cascade-form realizations for this system.
 - Sketch roughly the magnitude response $|H(\omega)|$ of this system.
- 7.9 Obtain the direct form I, direct form II, cascade, and parallel structures for the following systems.
- $y(n) = \frac{3}{4}y(n-1) - \frac{1}{8}y(n-2) + x(n) + \frac{1}{3}x(n-1)$
 - $y(n) = -0.1y(n-1) + 0.72y(n-2) + 0.7x(n) - 0.252x(n-2)$
 - $y(n) = -0.1y(n-1) + 0.2y(n-2) + 3x(n) + 3.6x(n-1) + 0.6x(n-2)$
 - $H(z) = \frac{2(1-z^{-1})(1+\sqrt{2}z^{-1}+z^{-2})}{(1+0.5z^{-1})(1-0.9z^{-1}+0.81z^{-2})}$

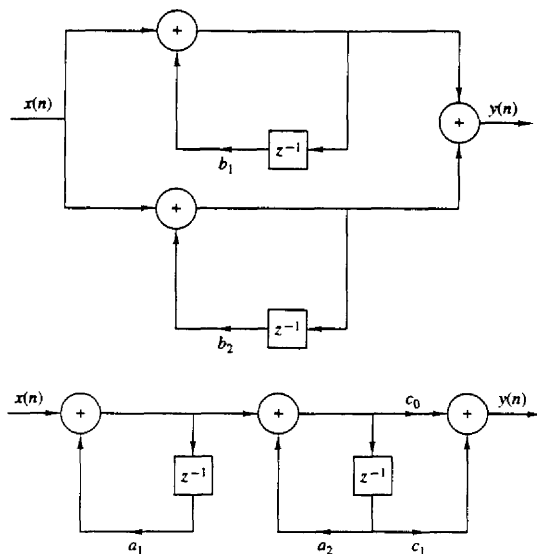


Figure P7.6

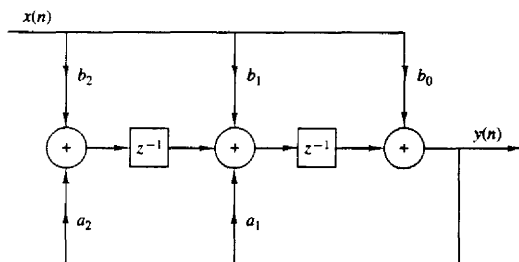


Figure P7.7

(e) $y(n) = \frac{1}{2}y(n-1) + \frac{1}{4}y(n-2) + x(n) + x(n-1)$

(f) $y(n) = y(n-1) - \frac{1}{2}y(n-2) + x(n) - x(n-1) + x(n-2)$

Which of the systems above are stable?

7.10 Show that the systems in Fig. P7.10 are equivalent.

7.11 Determine all the FIR filters which are specified by the lattice parameters $K_1 = \frac{1}{2}$, $K_2 = 0.6$, $K_3 = -0.7$, and $K_4 = \frac{1}{3}$.

7.12 Determine the set of difference equations for describing a realization of an IIR system based on the use of the transposed direct form II structure for the second-order subsystems.

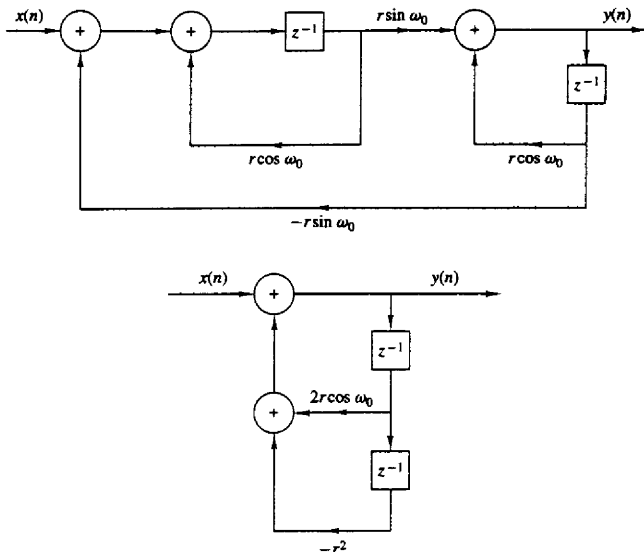


Figure P7.10

- 7.13* Write a program that implements a parallel-form realization based on transposed direct form II second-order modules.
- 7.14* Write a program that implements a cascade-form realization based on regular direct form II second-order modules.
- 7.15 Determine the parameters $\{K_m\}$ of the lattice filter corresponding to the FIR filter described by the system function

$$H(z) = A_2(z) = 1 + 2z^{-1} + z^{-2}$$

- 7.16 (a) Determine the zeros and sketch the zero pattern for the FIR lattice filter with parameters

$$K_1 = \frac{1}{2}, \quad K_2 = -\frac{1}{3}, \quad K_3 = 1$$

- (b) The same as in part (a) but with $K_3 = -1$.
- (c) You should have found that all the zeros lie exactly on the unit circle. Can this result be generalized? How?
- (d) Sketch the phase response of the filters in parts (a) and (b). What did you notice? Can this result be generalized? How?
- 7.17 Consider an FIR lattice filter with coefficients $K_1 = 0.65$, $K_2 = -0.34$, and $K_3 = 0.8$.
- (a) Find its impulse response by tracing a unit impulse input through the lattice structure.
- (b) Draw the equivalent direct-form structure.

7.18 Consider a causal IIR system with system function

$$H(z) = \frac{1 + 2z^{-1} + 3z^{-2} + 2z^{-3}}{1 + 0.9z^{-1} - 0.8z^{-2} + 0.5z^{-3}}$$

(a) Determine the equivalent lattice-ladder structure.

(b) Check if the system is stable.

7.19 Determine the input-output relationship, the system function, and plot the pole-zero pattern for the discrete-time system shown in Fig. P7.19.

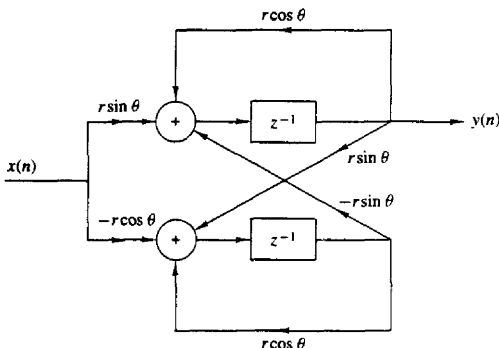


Figure P7.19

7.20 Determine the coupled-form state-space realization for the digital resonator

$$H(z) = \frac{1}{1 - (2r \cos \omega_0)z^{-1} + r^2 z^{-2}}$$

7.21 (a) Determine the impulse response of an FIR lattice filter with parameters $K_1 = 0.6$, $K_2 = 0.3$, $K_3 = 0.5$, and $K_4 = 0.9$.

(b) Sketch the direct form and lattice all-zero and all-pole filters specified by the K -parameters given in part (a).

7.22 (a) Sketch the lattice realization for the resonator

$$H(z) = \frac{1}{1 - (2r \cos \omega_0)z^{-1} + r^2 z^{-2}}$$

(b) What happens if $r = 1$?

7.23 Sketch the lattice-ladder structure for the system

$$H(z) = \frac{1 - 0.8z^{-1} + 0.15z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

7.24 Determine a state-space model and the corresponding realization for the following FIR system:

$$y(n) = \sum_{k=0}^M b_k x(n-k)$$

7.25 Determine the state-space model for the system described by

$$y(n) = y(n-1) + 0.11y(n-2) + x(n)$$

and sketch the type 1 and type 2 state-space realizations.

7.26 Determine the type 1 and type 2 state-space realizations for the Fibonacci system and its diagonal form.

7.27 By means of the z -transform, determine the impulse response of the system described by the state-space parameters

$$\mathbf{F} = \begin{bmatrix} 0 & 0.11 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} 0.11 \\ 1 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad d = 1$$

7.28 Determine the characteristic polynomial of the coupled-form state-space structure described by (7.4.68) and solve for the roots.

7.29 Determine the transpose structure for the coupled-form state-space structure shown in Fig. 7.31.

7.30 Consider a pole-zero system with system function

$$H(z) = \frac{(1 - 0.5e^{j\pi/4}z^{-1})(1 - 0.5e^{-j\pi/4}z^{-1})}{(1 - 0.8e^{j\pi/3}z^{-1})(1 - 0.8e^{-j\pi/3}z^{-1})}$$

- Sketch the regular and transpose direct form II realizations of the system.
- Determine and sketch the type 1 and type 2 state-space realizations.
- Determine the impulse response of the system by inverting $H(z)$ and by using state-space techniques.
- Determine the coupled-form state-space realization.
- Repeat parts (a) through (d) for the system obtained by changing the angle of the poles from $\pi/3$ to $\pi/4$.

7.31 (a) Determine a parallel and a cascade realization of the system

$$H(z) = \frac{1 + z^{-1}}{(1 - z^{-1})(1 - 0.8e^{j\pi/4}z^{-1})(1 - 0.8e^{-j\pi/4}z^{-1})}$$

- Determine the type 1 and type 2 state-space descriptions of the system in part (a).

7.32 Show how to use a lattice structure to implement the following all-pass filter

$$H(z) = \frac{0.5 + 0.2z^{-1} - 0.6z^{-2} + z^{-3}}{1 - 0.6z^{-1} + 0.2z^{-2} + 0.5z^{-3}}$$

Is the system stable?

7.33 Consider a system described by the following state-space equations:

$$\begin{aligned} \mathbf{v}(n+1) &= \begin{bmatrix} 0 & 1 \\ -0.81 & 1 \end{bmatrix} \mathbf{v}(n) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} x(n) \\ y(n) &= [-1.81 \quad 1] \mathbf{v}(n) + x(n) \end{aligned}$$

- Determine the characteristic polynomial and the eigenvalues of the system.
- Determine the state transition matrix $\Phi(n)$ for $n \geq 0$.
- Determine the system function and the impulse response of the system.
- Compute the step response of the system if $\mathbf{v}(0) = [0 \quad 1]^T$.
- Sketch a state-space realization for the system.

7.34 Repeat Problem 7.33 if the system is described by the state-space equations

$$\begin{aligned}\mathbf{v}(n+1) &= \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{v}(n) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} x(n) \\ y(n) &= [1 \quad 0] \mathbf{v}(n)\end{aligned}$$

7.35 Repeat Problem 7.33 for the system described by the state-space equations

$$\begin{aligned}\mathbf{v}(n+1) &= \begin{bmatrix} -0.3 & 0.4 \\ 0.4 & -0.3 \end{bmatrix} \mathbf{v}(n) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} x(n) \\ y(n) &= [1 \quad 1] \mathbf{v}(n) + x(n)\end{aligned}$$

7.36 Consider the system

$$y(n) = 0.9y(n-1) - 0.08y(n-2) + x(n) + x(n-1)$$

- Determine the type 1 and type 2 state-space realizations of the system.
- Determine the parallel and cascade state-space realizations of the system.
- Determine the impulse response of the system by at least two different methods.

7.37 Consider the causal system

$$y(n) = \frac{3}{4}y(n-1) - \frac{1}{8}y(n-2) + x(n) + \frac{1}{3}x(n-1)$$

- Determine its system function.
- Determine the type 1 state-space model.
- Determine the state transition matrix $\Phi(n) = \mathbf{F}^n$, for any n , using z -transform techniques.
- Determine the system function using the formula

$$H(z) = \mathbf{g}'(z\mathbf{I} - \mathbf{F})^{-1}\mathbf{q} + d$$

Compare the answer with that in part (a).

- Compute the characteristic polynomial $\det(z\mathbf{I} - \mathbf{F})$ and check if the system is stable.

7.38 Determine the impulse response of the system

$$\mathbf{F} = \begin{bmatrix} 0 & 0.11 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} 0.11 \\ 1 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad d = 1$$

using the z -transform approach.

7.39 A discrete-time system is described by the following state-space model:

$$\mathbf{v}(n+1) = \mathbf{F}\mathbf{v}(n) + \mathbf{q}x(n)$$

$$y(n) = \mathbf{g}'\mathbf{v}(n) + dx(n)$$

where

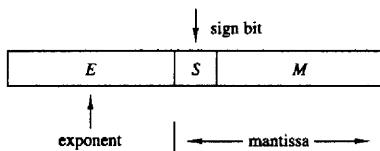
$$\mathbf{F} = \begin{bmatrix} 0 & 1 \\ -\frac{5}{16} & -1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} \frac{11}{8} \\ 2 \end{bmatrix}, \quad d = 2$$

- Sketch the corresponding state-space structure.
- Calculate the impulse response for $n = 0, 1, \dots, 5$ and for $n = 17$ by using the state-space approach
- Find the difference equation description of the system.
- Repeat part (b) by using the difference equation.
- Sketch the direct form II implementation of the system.

7.40 Determine the state-space parameters F , q , g , and d for:

- the all-zero lattice structure
- the all-pole lattice structure

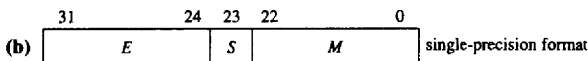
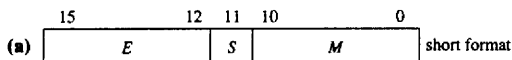
7.41 The generic floating-point format for a DSP microprocessor is the following:



The value of the number X is given by

$$X = \begin{cases} 01.M \times 2^E & \text{if } S = 0 \\ 10.M \times 2^E & \text{if } S = 1 \\ 0 & \text{if } E \text{ is the most negative two's-complement value} \end{cases}$$

Determine the range of positive and negative numbers for the following two formats:



7.42 Consider the IIR recursive filter shown in Fig. P7.42 and let $h_F(n)$, $h_R(n)$, and $h(n)$ denote the impulse responses of the FIR section, the recursive section, and the overall filter, respectively.

- Find all the causal and stable recursive second-order sections with integer coefficients (a_1 , a_2) and determine and sketch their impulse responses and frequency responses. These filters do not require complicated multiplications or quantization after multiplications.

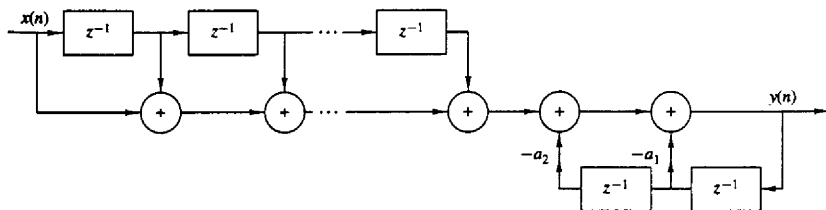


Figure P7.42

- (b) Show that three of the sections obtained in part (a) can be obtained by interconnection of other sections.
- (c) Find a difference equation that describes the impulse response $h(n)$ of the filter and determine the conditions for the overall filter to be FIR.
- (d) Rederive the results in parts (a) to (c) using z -domain considerations.

7.43 This problem illustrates the development of digital filter structures using Horner's rule for polynomial evaluation. To this end consider the polynomial

$$p(x) = \alpha_p x^p + \alpha_{p-1} x^{p-1} + \cdots + \alpha_1 x + \alpha_0$$

which computes $p(x)$ with the minimum cost of p multiplications and p additions.

- (a) Draw the structures corresponding to the factorizations

$$H_1(z) = b_0(1 + b_1 z^{-1}(1 + b_2 z^{-1}(1 + b_3 z^{-1})))$$

$$H(z) = b_0(z^{-3} + (b_1 z^{-2} + (b_2 z^{-1} + b_3)))$$

and determine the system function, number of delay elements, and arithmetic operations for each structure

- (b) Draw the Horner structure for the following linear-phase system:

$$H(z) = z^{-1} \left[\alpha_0 + \sum_{k=1}^3 (z^{-k} + z^k) \alpha_k \right]$$

7.44 Let x_1 and x_2 be $(b+1)$ -bit binary numbers with magnitude less than 1. To compute the sum of x_1 and x_2 using two's-complement representation we treat them as $(b+1)$ -bit unsigned numbers, we perform addition modulo-2 and ignore any carry after the sign bit.

- (a) Show that if the sum of two numbers with the same sign has the opposite sign, this corresponds to overflow.
- (b) Show that when we compute the sum of several numbers using two's-complement representation, the result will be correct, even if there are overflows, if the correct sum is less than 1 in magnitude. Illustrate this argument by constructing a simple example with three numbers.

7.45 Consider the system described by the difference equation

$$y(n) = ay(n-1) - ax(n) + x(n-1)$$

- (a) Show that it is all-pass.
- (b) Obtain the direct form II realization of the system
- (c) If you quantize the coefficients of the system in part (b), is it still all-pass?
- (d) Obtain a realization by rewriting the difference equation as

$$y(n) = a[y(n-1) - x(n)] + x(n-1)$$

- (e) If you quantize the coefficients of the system in part (d), is it still all-pass?

7.46 Consider the system

$$y(n) = \frac{1}{2}y(n-1) + x(n)$$

- (a) Compute its response to the input $x(n) = (\frac{1}{4})^n u(n)$ assuming infinite-precision arithmetic.

- (b) Compute the response of the system $y(n)$, $0 \leq n \leq 5$ to the same input, assuming finite-precision sign-and-magnitude fractional arithmetic with five bits (i.e., the sign bit plus four fractional bits). The quantization is performed by truncation.
- (c) Compare the results obtained in parts (a) and (b).

7.47 The input to the system

$$y(n) = 0.999y(n-1) + x(n)$$

is quantized to $b = 8$ bits. What is the power produced by the quantization noise at the output of the filter?

7.48 Consider the system

$$y(n) = 0.875y(n-1) - 0.125y(n-2) + x(n)$$

- (a) Compute its poles and design the cascade realization of the system.
- (b) Quantize the coefficients of the system using truncation, maintaining a sign bit plus three other bits. Determine the poles of the resulting system.
- (c) Repeat part (b) for the same precision using rounding.
- (d) Compare the poles obtained in parts (b) and (c) with those in part (a). Which realization is better? Sketch the frequency responses of the systems in parts (a), (b), and (c).

7.49 Consider the system

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 + \frac{1}{4}z^{-1})}$$

- (a) Draw all possible realizations of the system.
- (b) Suppose that we implement the filter with fixed-point sign-and-magnitude fractional arithmetic using $(b+1)$ bits (one bit is used for the sign). Each resulting product is rounded into b bits. Determine the variance of the round-off noise created by the multipliers at the output of each one of the realizations in part (a).
- 7.50** The first-order filter shown in Fig. P7.50 is implemented in four-bit (including sign) fixed-point two's-complement fractional arithmetic. Products are rounded to four-bit representation. Using the input $x(n) = 0.10\delta(n)$, determine:

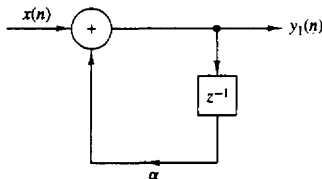


Figure P7.50

- (a) The first five outputs if $\alpha = 0.5$. Does the filter go into a limit cycle?
- (b) The first five outputs if $\alpha = 0.75$. Does the filter go into a limit cycle?
- 7.51** The digital system shown in Fig. P7.51 uses a six-bit (including sign) fixed-point two's-complement A/D converter with rounding, and the filter $H(z)$ is implemented using eight-bit (including sign) fixed-point two's-complement fractional arithmetic with rounding. The input $x(t)$ is a zero-mean uniformly distributed random process having autocorrelation $\gamma_{xx}(\tau) = 3\delta(\tau)$. Assume that the A/D converter can handle input values up to ± 1.0 without overflow.

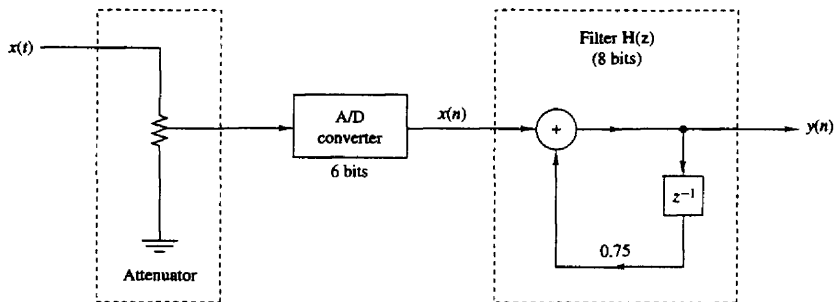


Figure P7.51

- (a) What value of attenuation should be applied prior to the A/D converter to assure that it does not overflow?
- (b) With the attenuation above, what is the signal-to-quantization noise ratio (SNR) at the A/D converter output?
- (c) The six-bit A/D samples can be left-justified, right-justified, or centered in the eight-bit word used as the input to the digital filter. What is the correct strategy to use for maximum SNR at the filter output without overflow?
- (d) What is the SNR at the output of the filter due to all quantization noise sources?
- 7.52** Shown in Fig. P7.52 is the coupled-form implementation of a two-pole filter with poles at $x = re^{\pm j\theta}$. There are four real multiplications per output point. Let $e_i(n)$, $i = 1, 2, 3, 4$ represent the round-off noise in a fixed-point implementation of the filter. Assume that the noise sources are zero-mean mutually uncorrelated stationary white noise sequences. For each n the probability density function $p(e)$ is uniform in the range $-\Delta/2 \leq e \leq \Delta/2$, where $\Delta = 2^{-b}$.
- (a) Write the two coupled difference equations for $y(n)$ and $v(n)$, including the noise sources and the input sequence $x(n)$.
- (b) From these two difference equations, show that the filter system functions $H_1(z)$ and $H_2(z)$ between the input noise terms $e_1(n) + e_2(n)$ and $e_3(n) + e_4(n)$ and the output $y(n)$ are:

$$H_1(z) = \frac{r \sin \theta z^{-1}}{1 - 2r \cos \theta z^{-1} + r^2 z^{-2}}$$

$$H_2(z) = \frac{1 - r \cos \theta z^{-1}}{1 - 2r \cos \theta z^{-1} + r^2 z^{-2}}$$

We know that

$$H(z) = \frac{1}{1 - 2r \cos \theta z^{-1} + r^2 z^{-2}} \Rightarrow h(n) = \frac{1}{\sin \theta} r^n \sin(n+1)\theta u(n)$$

Determine $h_1(n)$ and $h_2(n)$.

- (c) Determine a closed-form expression for the variance of the total noise from $e_i(n)$, $i = 1, 2, 3, 4$ at the output of the filter.

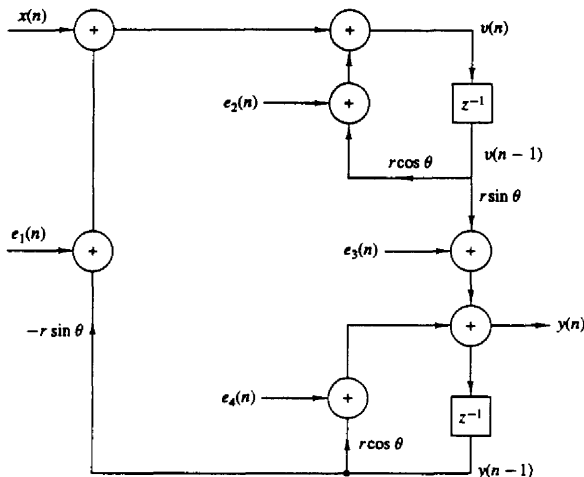


Figure P7.52

- 7.53 Determine the variance of the round-off noise at the output of the two cascade realizations of the filter shown in Fig. P7.53, with system function

$$H(z) = H_1(z)H_2(z)$$

where

$$H_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

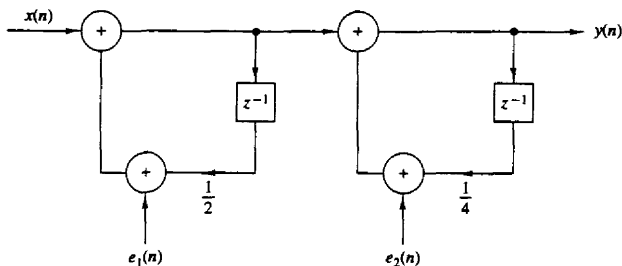
$$H_2(z) = \frac{1}{1 - \frac{1}{4}z^{-1}}$$

- 7.54 *Quantization effects in direct-form FIR filters* Consider a direct-form realization of an FIR filter of length M . Suppose that the multiplication of each coefficient with the corresponding signal sample is performed in fixed-point arithmetic with b bits and each product is rounded to b bits. Determine the variance of the quantization noise at the output of the filter by using a statistical characterization of the round-off noise as in Section 7.7.3.

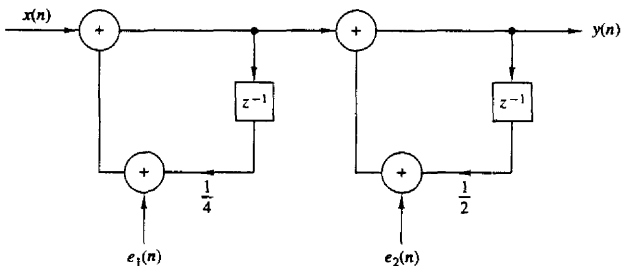
- 7.55* Consider the system specified by the system function

$$H(z) = \frac{B(z)}{A(z)} = \left[G_1 \frac{(1 - 0.8e^{j\pi/4}z^{-1})(1 - 0.8e^{-j\pi/4}z^{-1})}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{3}z^{-1})} \right] \left[G_2 \frac{(1 + \frac{1}{4}z^{-1})(1 - \frac{5}{8}z^{-1})}{(1 - 0.8e^{j\pi/3}z^{-1})(1 - 0.8e^{-j\pi/3}z^{-1})} \right]$$

- (a) Choose G_1 and G_2 so that the gain of each second-order section at $\omega = 0$ is equal to 1.



(a) Cascade realization I



(b) Cascade realization II

Figure P7.53

- (b) Sketch the direct form 1, direct form 2, and cascade realizations of the system.
 (c) Write a program that implements the direct form 1 and direct form 2, and compute the first 100 samples of the impulse response and the step response of the system.
 (d) Plot the results in part (c) to illustrate the proper functioning of the programs.

7.56* Consider the system given in Problem 7.55 with $G_1 = G_2 = 1$.

- (a) Determine a lattice realization for the system

$$H(z) = B(z)$$

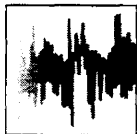
- (b) Determine a lattice realization for the system

$$H(z) = \frac{1}{A(z)}$$

- (c) Determine a lattice-ladder realization for the system $H(z) = B(z)/A(z)$.
 (d) Write a program for the implementation of the lattice-ladder structure in part (c).
 (e) Determine and sketch the first 100 samples of the impulse responses of the systems in parts (a) through (c) by working with the lattice structures.
 (f) Compute and sketch the first 100 samples of the convolution of impulse responses in parts (a) and (b). What did you find? Explain your results.

7.57* Consider the system given in Problem 7.55.

- (a) Determine the parallel-form structure and write a program for its implementation.
- (b) Sketch a parallel structure using second-order coupled-form state-space sections.
- (c) Write a program for the implementation of the structure in part (b).
- (d) Verify the programs in parts (a) and (c) by computing and sketching the impulse response of the system.



8

Design of Digital Filters

With the background that we have developed in the preceding chapters, we are now in a position to treat the subject of digital filter design. We shall describe several methods for designing FIR and IIR digital filters.

In the design of frequency-selective filters, the desired filter characteristics are specified in the frequency domain in terms of the desired magnitude and phase response of the filter. In the filter design process, we determine the coefficients of a causal FIR or IIR filter that closely approximates the desired frequency response specifications. The issue of which type of filter to design, FIR or IIR, depends on the nature of the problem and on the specifications of the desired frequency response.

In practice, FIR filters are employed in filtering problems where there is a requirement for a linear-phase characteristic within the passband of the filter. If there is no requirement for a linear-phase characteristic, either an IIR or an FIR filter may be employed. However, as a general rule, an IIR filter has lower sidelobes in the stopband than an FIR filter having the same number of parameters. For this reason, if some phase distortion is either tolerable or unimportant, an IIR filter is preferable, primarily because its implementation involves fewer parameters, requires less memory and has lower computational complexity.

In conjunction with our discussion of digital filter design, we describe frequency transformations in both the analog and digital domains for transforming a lowpass prototype filter into either another lowpass, bandpass, bandstop, or high-pass filter.

Today, FIR and IIR digital filter design is greatly facilitated by the availability of numerous computer software programs. In describing the various digital filter design methods in this chapter, our primary objective is to give the reader the background necessary to select the filter that best matches the application and satisfies the design requirements.

8.1 GENERAL CONSIDERATIONS

In Section 4.5, we described the characteristics of ideal filters and demonstrated that such filters are not causal and therefore, are not physically realizable. In this

section, the issue of causality and its implications is considered in more detail. Following this discussion, we present the frequency response characteristics of causal FIR and IIR digital filters.

8.1.1 Causality and Its Implications

Let us consider the issue of causality in more detail by examining the impulse response $h(n)$ of an ideal lowpass filter with frequency response characteristic

$$H(\omega) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & \omega_c < \omega \leq \pi \end{cases} \quad (8.1.1)$$

The impulse response of this filter is

$$h(n) = \begin{cases} \frac{\omega_c}{\pi}, & n = 0 \\ \frac{\omega_c}{\pi} \frac{\sin \omega_c n}{\omega_c n}, & n \neq 0 \end{cases} \quad (8.1.2)$$

A plot of $h(n)$ for $\omega_c = \pi/4$ is illustrated in Fig. 8.1. It is clear that the ideal lowpass filter is noncausal and hence it cannot be realized in practice.

One possible solution is to introduce a large delay n_0 in $h(n)$ and arbitrarily to set $h(n) = 0$ for $n < n_0$. However, the resulting system no longer has an ideal frequency response characteristic. Indeed, if we set $h(n) = 0$ for $n < n_0$, the Fourier series expansion of $H(\omega)$ results in the Gibbs phenomenon, as will be described in Section 8.2.

Although this discussion is limited to the realization of a lowpass filter, our conclusions hold, in general, for all the other ideal filter characteristics. In brief, none of the ideal filter characteristics previously illustrated in Fig. 4.43 are causal, hence all are physically unrealizable.

A question that naturally arises at this point is the following: What are the necessary and sufficient conditions that a frequency response characteristic $H(\omega)$

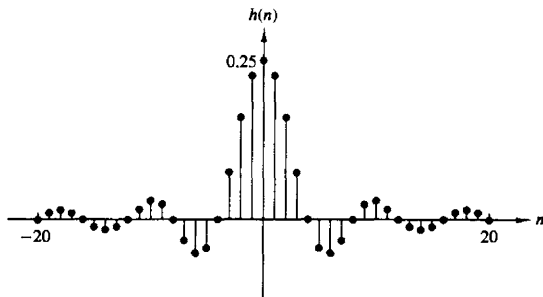


Figure 8.1 Unit sample response of an ideal lowpass filter.

must satisfy in order for the resulting filter to be causal? The answer to this question is given by the Paley–Wiener theorem, which can be stated as follows:

Paley–Wiener Theorem. If $h(n)$ has finite energy and $h(n) = 0$ for $n < 0$, then [for a reference, see Wiener and Paley (1934)]

$$\int_{-\pi}^{\pi} |\ln |H(\omega)|| d\omega < \infty \quad (8.1.3)$$

Conversely, if $|H(\omega)|$ is square integrable and if the integral in (8.1.3) is finite, then we can associate with $|H(\omega)|$ a phase response $\Theta(\omega)$, so that the resulting filter with frequency response

$$H(\omega) = |H(\omega)|e^{j\Theta(\omega)}$$

is causal.

One important conclusion that we draw from the Paley–Wiener theorem is that the magnitude function $|H(\omega)|$ can be zero at some frequencies, but it cannot be zero over any finite band of frequencies, since the integral then becomes infinite. Consequently, any ideal filter is noncausal.

Apparently, causality imposes some tight constraints on a linear time-invariant system. In addition to the Paley–Wiener condition, causality also implies a strong relationship between $H_R(\omega)$ and $H_I(\omega)$, the real and imaginary components of the frequency response $H(\omega)$. To illustrate this dependence, we decompose $h(n)$ into an even and an odd sequence, that is,

$$h(n) = h_e(n) + h_o(n) \quad (8.1.4)$$

where

$$h_e(n) = \frac{1}{2}[h(n) + h(-n)] \quad (8.1.5)$$

and

$$h_o(n) = \frac{1}{2}[h(n) - h(-n)] \quad (8.1.6)$$

Now, if $h(n)$ is causal, it is possible to recover $h(n)$ from its even part $h_e(n)$ for $0 \leq n \leq \infty$ or from its odd component $h_o(n)$ for $1 \leq n \leq \infty$.

Indeed, it can be easily seen that

$$h(n) = 2h_e(n)u(n) - h_e(0)\delta(n) \quad n \geq 0 \quad (8.1.7)$$

and

$$h(n) = 2h_o(n)u(n) + h(0)\delta(n) \quad n \geq 1 \quad (8.1.8)$$

Since $h_o(n) = 0$ for $n = 0$, we cannot recover $h(0)$ from $h_o(n)$ and hence we also must know $h(0)$. In any case, it is apparent that $h_o(n) = h_e(n)$ for $n \geq 1$, so there is a strong relationship between $h_o(n)$ and $h_e(n)$.

If $h(n)$ is absolutely summable (i.e., BIBO stable), the frequency response $H(\omega)$ exists, and

$$H(\omega) = H_R(\omega) + jH_I(\omega) \quad (8.1.9)$$

In addition, if $h(n)$ is real valued and causal, the symmetry properties of the Fourier transform imply that

$$\begin{aligned} h_e(n) &\xleftrightarrow{F} H_R(\omega) \\ h_o(n) &\xleftrightarrow{F} H_I(\omega) \end{aligned} \quad (8.1.10)$$

Since $h(n)$ is completely specified by $h_e(n)$, it follows that $H(\omega)$ is completely determined if we know $H_R(\omega)$. Alternatively, $H(\omega)$ is completely determined from $H_I(\omega)$ and $h(0)$. In short, $H_R(\omega)$ and $H_I(\omega)$ are interdependent and cannot be specified independently if the system is causal. Equivalently, the magnitude and phase responses of a causal filter are interdependent and hence cannot be specified independently.

Given $H_R(\omega)$ for a corresponding real, even, and absolutely summable sequence $h_e(n)$, we can determine $H(\omega)$. The following example illustrates the procedure.

Example 8.1.1

Consider a stable LTI system with real and even impulse response $h(n)$. Determine $H(\omega)$ if

$$H_R(\omega) = \frac{1 - a \cos \omega}{1 - 2a \cos \omega + a^2}, \quad |a| < 1$$

Solution The first step is to determine $h_e(n)$. This can be done by noting that

$$H_R(\omega) = H_R(z)|_{z=e^{j\omega}}$$

where

$$H_R(z) = \frac{1 - a(z + z^{-1})/2}{1 - a(z + z^{-1}) + a^2} = \frac{z - a(z^2 + 1)/2}{(z - a)(1 - az)}$$

The ROC has to be restricted by the poles at $p_1 = a$ and $p_2 = 1/a$ and should include the unit circle. Hence the ROC is $|a| < |z| < 1/|a|$. Consequently, $h_e(n)$ is a two-sided sequence, with the pole at $z = a$ contributing to the causal part and $p_2 = 1/a$ contributing to the anticausal part. By using a partial-fraction expansion, we obtain

$$h_e(n) = \frac{1}{2}a^{|n|} + \frac{1}{2}\delta(n) \quad (8.1.11)$$

By substituting (8.1.11) into (8.1.7), we obtain $h(n)$ as

$$h(n) = a^{|n|}u(n)$$

Finally, we obtain the Fourier transform of $h(n)$ as

$$H(\omega) = \frac{1}{1 - ae^{-j\omega}}$$

The relationship between the real and imaginary components of the Fourier transform of an absolutely summable, causal, and real sequence can be easily established from (8.1.7). The Fourier transform relationship for (8.1.7) is

$$H(\omega) = H_R(\omega) + jH_I(\omega) = \frac{1}{\pi} \int_{-\pi}^{\pi} H_R(\lambda) U(\omega - \lambda) d\lambda - h_e(0) \quad (8.1.12)$$

where $U(\omega)$ is the Fourier transform of the unit step sequence $u(n)$. Although the unit step sequence is not absolutely summable, it has a Fourier transform (see Section 4.2.8).

$$\begin{aligned} U(\omega) &= \pi \delta(\omega) + \frac{1}{1 - e^{-j\omega}} \\ &= \pi \delta(\omega) + \frac{1}{2} - j \frac{1}{2} \cot \frac{\omega}{2} \quad -\pi \leq \omega \leq \pi \end{aligned} \quad (8.1.13)$$

By substituting (8.1.13) into (8.1.12) and carrying out the integration, we obtain the relation between $H_R(\omega)$ and $H_I(\omega)$ as

$$H_I(\omega) = -\frac{1}{2\pi} \int_{-\pi}^{\pi} H_R(\lambda) \cot \frac{\omega - \lambda}{2} d\lambda \quad (8.1.14)$$

Thus $H_I(\omega)$ is uniquely determined from $H_R(\omega)$ through this integral relationship. The integral is called a *discrete Hilbert transform*. It is left as an exercise to the reader to establish the relationship for $H_R(\omega)$ in terms of the discrete Hilbert transform of $H_I(\omega)$.

To summarize, causality has very important implications in the design of frequency-selective filters. These are: (a) the frequency response $H(\omega)$ cannot be zero, except at a finite set of points in frequency; (b) the magnitude $|H(\omega)|$ cannot be constant in any finite range of frequencies and the transition from passband to stopband cannot be infinitely sharp [this is a consequence of the Gibbs phenomenon, which results from the truncation of $h(n)$ to achieve causality]; and (c) the real and imaginary parts of $H(\omega)$ are interdependent and are related by the discrete Hilbert transform. As a consequence, the magnitude $|H(\omega)|$ and phase $\Theta(\omega)$ of $H(\omega)$ cannot be chosen arbitrarily.

Now that we know the restrictions that causality imposes on the frequency response characteristic and the fact that ideal filters are not achievable in practice, we limit our attention to the class of linear time-invariant systems specified by the difference equation

$$y(n) = -\sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

which are causal and physically realizable. As we have demonstrated, such systems have a frequency response

$$H(\omega) = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{1 + \sum_{k=1}^N a_k e^{-j\omega k}} \quad (8.1.15)$$

The basic digital filter design problem is to approximate any of the ideal frequency response characteristics with a system that has the frequency response (8.1.15), by properly selecting the coefficients $\{a_k\}$ and $\{b_k\}$. The approximation problem is

treated in detail in Sections 8.2 and 8.3, where we discuss techniques for digital filter design.

8.1.2 Characteristics of Practical Frequency-Selective Filters

As we observed from our discussion of the preceding section, ideal filters are noncausal and hence physically unrealizable for real-time signal processing applications. Causality implies that the frequency response characteristic $H(\omega)$ of the filter cannot be zero, except at a finite set of points in the frequency range. In addition, $H(\omega)$ cannot have an infinitely sharp cutoff from passband to stopband, that is, $H(\omega)$ cannot drop from unity to zero abruptly.

Although the frequency response characteristics possessed by ideal filters may be desirable, they are not absolutely necessary in most practical applications. If we relax these conditions, it is possible to realize causal filters that approximate the ideal filters as closely as we desire. In particular, it is not necessary to insist that the magnitude $|H(\omega)|$ be constant in the entire passband of the filter. A small amount of ripple in the passband, as illustrated in Fig. 8.2, is usually tolerable. Similarly, it is not necessary for the filter response $|H(\omega)|$ to be zero in the stopband. A small, nonzero value or a small amount of ripple in the stopband is also tolerable.

The transition of the frequency response from passband to stopband defines the *transition band* or *transition region* of the filter, as illustrated in Fig. 8.2. The band-edge frequency ω_p defines the edge of the passband, while the frequency ω_s denotes the beginning of the stopband. Thus the width of the transition band is $\omega_s - \omega_p$. The width of the passband is usually called the *bandwidth* of the filter. For example, if the filter is lowpass with a passband edge frequency ω_p , its bandwidth is ω_p .

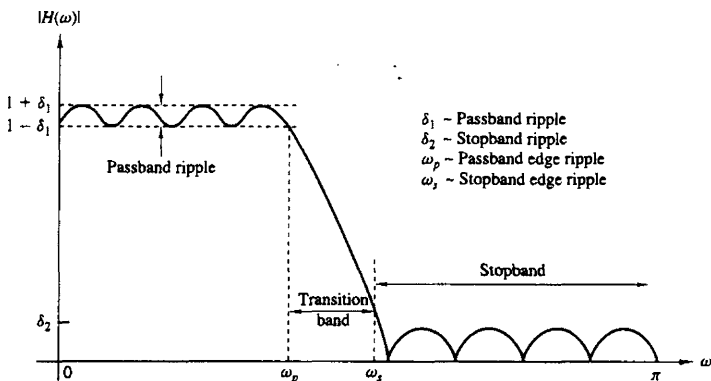


Figure 8.2 Magnitude characteristics of physically realizable filters.

If there is ripple in the passband of the filter, its value is denoted as δ_1 , and the magnitude $|H(\omega)|$ varies between the limits $1 \pm \delta_1$. The ripple in the stopband of the filter is denoted as δ_2 .

To accommodate a large dynamic range in the graph of the frequency response of any filter, it is common practice to use a logarithmic scale for the magnitude $|H(\omega)|$. Consequently, the ripple in the passband is $20 \log_{10} \delta_1$ decibels, and that in the stopband is $20 \log_{10} \delta_2$.

In any filter design problem we can specify (1) the maximum tolerable passband ripple, (2) the maximum tolerable stopband ripple, (3) the passband edge frequency ω_p , and (4) the stopband edge frequency ω_s . Based on these specifications, we can select the parameters $\{a_k\}$ and $\{b_k\}$ in the frequency response characteristic, given by (8.1.15), which best approximates the desired specification. The degree to which $H(\omega)$ approximates the specifications depends in part on the criterion used in the selection of the filter coefficients $\{a_k\}$ and $\{b_k\}$ as well as on the numbers (M, N) of coefficients.

In the following section we present a method for designing linear-phase FIR filters.

8.2 DESIGN OF FIR FILTERS

In this section we describe several methods for designing FIR filters. Our treatment is focused on the important class of linear-phase FIR filters.

8.2.1 Symmetric and Antisymmetric FIR Filters

An FIR filter of length M with input $x(n)$ and output $y(n)$ is described by the difference equation

$$\begin{aligned} y(n) &= b_0 x(n) + b_1 x(n-1) + \cdots + b_{M-1} x(n-M+1) \\ &= \sum_{k=0}^{M-1} b_k x(n-k) \end{aligned} \quad (8.2.1)$$

where $\{b_k\}$ is the set of filter coefficients. Alternatively, we can express the output sequence as the convolution of the unit sample response $h(n)$ of the system with the input signal. Thus we have

$$y(n) = \sum_{k=0}^{M-1} h(k) x(n-k) \quad (8.2.2)$$

where the lower and upper limits on the convolution sum reflect the causality and finite-duration characteristics of the filter. Clearly, (8.2.1) and (8.2.2) are identical in form and hence it follows that $b_k = h(k)$, $k = 0, 1, \dots, M-1$.

The filter can also be characterized by its system function

$$H(z) = \sum_{k=0}^{M-1} h(k) z^{-k} \quad (8.2.3)$$

which we view as a polynomial of degree $M - 1$ in the variable z^{-1} . The roots of this polynomial constitute the zeros of the filter.

An FIR filter has linear phase if its unit sample response satisfies the condition

$$h(n) = \pm h(M - 1 - n) \quad n = 0, 1, \dots, M - 1 \quad (8.2.4)$$

When the symmetry and antisymmetry conditions in (8.2.4) are incorporated into (8.2.3), we have

$$\begin{aligned} H(z) &= h(0) + h(1)z^{-1} + h(2)z^{-2} + \dots + h(M-2)z^{-(M-2)} + h(M-1)z^{-(M-1)} \\ &= z^{-(M-1)/2} \left\{ h\left(\frac{M-1}{2}\right) + \sum_{n=0}^{(M-3)/2} h(n) [z^{(M-1-2k)/2} \pm z^{-(M-1-2k)/2}] \right\} \quad M \text{ odd} \\ &= z^{-(M-1)/2} \sum_{n=0}^{(M/2)-1} h(n) [z^{(M-1-2k)/2} \pm z^{-(M-1-2k)/2}] \quad M \text{ even} \end{aligned} \quad (8.2.5)$$

Now, if we substitute z^{-1} for z in (8.2.3) and multiply both sides of the resulting equation by $z^{-(M-1)}$, we obtain

$$z^{-(M-1)} H(z^{-1}) = \pm H(z) \quad (8.2.6)$$

This result implies that the roots of the polynomial $H(z)$ are identical to the roots of the polynomial $H(z^{-1})$. Consequently, the roots of $H(z)$ must occur in reciprocal pairs. In other words, if z_1 is a root or a zero of $H(z)$, then $1/z_1$ is also a root. Furthermore, if the unit sample response $h(n)$ of the filter is real, complex-valued roots must occur in complex-conjugate pairs. Hence, if z_1 is a complex-valued root, z_1^* is also a root. As a consequence of (8.2.6), $H(z)$ also has a zero at $1/z_1^*$. Figure 8.3 illustrates the symmetry that exists in the location of the zeros of a linear-phase FIR filter.

The frequency response characteristics of linear-phase FIR filters are obtained by evaluating (8.2.5) on the unit circle. This substitution yields the expression for $H(\omega)$.

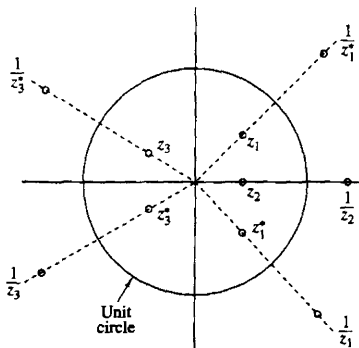


Figure 8.3 Symmetry of zero locations for a linear-phase FIR filter.

When $h(n) = h(M-1-n)$, $H(\omega)$ can be expressed as

$$H(\omega) = H_r(\omega)e^{-j\omega(M-1)/2} \quad (8.2.7)$$

where $H_r(\omega)$ is a real function of ω and can be expressed as

$$H_r(\omega) = h\left(\frac{M-1}{2}\right) + 2 \sum_{n=0}^{(M-3)/2} h(n) \cos \omega \left(\frac{M-1}{2} - n\right) \quad M \text{ odd} \quad (8.2.8)$$

$$H_r(\omega) = 2 \sum_{n=0}^{(M/2)-1} h(n) \cos \omega \left(\frac{M-1}{2} - n\right) \quad M \text{ even} \quad (8.2.9)$$

The phase characteristic of the filter for both M odd and M even is

$$\Theta(\omega) = \begin{cases} -\omega \left(\frac{M-1}{2}\right), & \text{if } H_r(\omega) > 0 \\ -\omega \left(\frac{M-1}{2}\right) + \pi, & \text{if } H_r(\omega) < 0 \end{cases} \quad (8.2.10)$$

When

$$h(n) = -h(M-1-n)$$

the unit sample response is *antisymmetric*. For M odd, the center point of the antisymmetric $h(n)$ is $n = (M-1)/2$. Consequently,

$$h\left(\frac{M-1}{2}\right) = 0$$

However, if M is even, each term in $h(n)$ has a matching term of opposite sign.

It is straightforward to show that the frequency response of an FIR filter with an antisymmetric unit sample response can be expressed as

$$H(\omega) = H_r(\omega)e^{j[-\omega(M-1)/2 + \pi/2]} \quad (8.2.11)$$

where

$$H_r(\omega) = 2 \sum_{n=0}^{(M-3)/2} h(n) \sin \omega \left(\frac{M-1}{2} - n\right) \quad M \text{ odd} \quad (8.2.12)$$

$$H_r(\omega) = 2 \sum_{n=0}^{(M/2)-1} h(n) \sin \omega \left(\frac{M-1}{2} - n\right) \quad M \text{ even} \quad (8.2.13)$$

The phase characteristic of the filter for both M odd and M even is

$$\Theta(\omega) = \begin{cases} \frac{\pi}{2} - \omega \left(\frac{M-1}{2}\right), & \text{if } H_r(\omega) > 0 \\ \frac{3\pi}{2} - \omega \left(\frac{M-1}{2}\right), & \text{if } H_r(\omega) < 0 \end{cases} \quad (8.2.14)$$

These general frequency response formulas can be used to design linear-phase FIR filters with symmetric and antisymmetric unit sample responses. We

note that, for a symmetric $h(n)$, the number of filter coefficients that specify the frequency response is $(M+1)/2$ when M is odd or $M/2$ when M is even. On the other hand, if the unit sample response is antisymmetric,

$$h\left(\frac{M-1}{2}\right) = 0$$

so that there are $(M-1)/2$ filter coefficients when M is odd and $M/2$ coefficients when M is even to be specified.

The choice of a symmetric or antisymmetric unit sample response depends on the application. As we shall see later, a symmetric unit sample response is suitable for some applications, while an antisymmetric unit sample response is more suitable for other applications. For example, if $h(n) = -h(M-1-n)$ and M is odd, (8.2.12) implies that $H_r(0) = 0$ and $H_r(\pi) = 0$. Consequently, (8.2.12) is not suitable as either a lowpass filter or a highpass filter. Similarly, the antisymmetric unit sample response with M even also results in $H_r(0) = 0$, as can be easily verified from (8.2.13). Consequently, we would not use the antisymmetric condition in the design of a lowpass linear-phase FIR filter. On the other hand, the symmetry condition $h(n) = h(M-1-n)$ yields a linear-phase FIR filter with a nonzero response at $\omega = 0$, if desired, that is,

$$H_r(0) = h\left(\frac{M-1}{2}\right) + 2 \sum_{n=0}^{(M-3)/2} h(n), \quad M \text{ odd} \quad (8.2.15)$$

$$H_r(0) = 2 \sum_{n=0}^{(M/2)-1} h(n), \quad M \text{ even} \quad (8.2.16)$$

In summary, the problem of FIR filter design is simply to determine the M coefficients $h(n)$, $n = 0, 1, \dots, M-1$, from a specification of the desired frequency response $H_d(\omega)$ of the FIR filter. The important parameters in the specification of $H_d(\omega)$ are given in Fig. 8.2.

In the following subsections we describe design methods based on specification of $H_d(\omega)$.

8.2.2 Design of Linear-Phase FIR Filters Using Windows

In this method we begin with the desired frequency response specification $H_d(\omega)$ and determine the corresponding unit sample response $h_d(n)$. Indeed, $h_d(n)$ is related to $H_d(\omega)$ by the Fourier transform relation

$$H_d(\omega) = \sum_{n=0}^{\infty} h_d(n) e^{-j\omega n} \quad (8.2.17)$$

where

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \quad (8.2.18)$$

Thus, given $H_d(\omega)$, we can determine the unit sample response $h_d(n)$ by evaluating the integral in (8.2.18).

In general, the unit sample response $h_d(n)$ obtained from (8.2.17) is infinite in duration and must be truncated at some point, say at $n = M - 1$, to yield an FIR filter of length M . Truncation of $h_d(n)$ to a length $M - 1$ is equivalent to multiplying $h_d(n)$ by a "rectangular window," defined as

$$w(n) = \begin{cases} 1, & n = 0, 1, \dots, M - 1 \\ 0, & \text{otherwise} \end{cases} \quad (8.2.19)$$

Thus the unit sample response of the FIR filter becomes

$$\begin{aligned} h(n) &= h_d(n)w(n) \\ &= \begin{cases} h_d(n), & n = 0, 1, \dots, M - 1 \\ 0, & \text{otherwise} \end{cases} \end{aligned} \quad (8.2.20)$$

It is instructive to consider the effect of the window function on the desired frequency response $H_d(\omega)$. Recall that multiplication of the window function $w(n)$ with $h_d(n)$ is equivalent to convolution of $H_d(\omega)$ with $W(\omega)$, where $W(\omega)$ is the frequency-domain representation (Fourier transform) of the window function, that is,

$$W(\omega) = \sum_{n=0}^{M-1} w(n)e^{-j\omega n} \quad (8.2.21)$$

Thus the convolution of $H_d(\omega)$ with $W(\omega)$ yields the frequency response of the (truncated) FIR filter. That is,

$$H(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\nu) W(\omega - \nu) d\nu \quad (8.2.22)$$

The Fourier transform of the rectangular window is

$$\begin{aligned} W(\omega) &= \sum_{n=0}^{M-1} e^{-j\omega n} \\ &= \frac{1 - e^{-j\omega M}}{1 - e^{-j\omega}} = e^{-j\omega(M-1)/2} \frac{\sin(\omega M/2)}{\sin(\omega/2)} \end{aligned} \quad (8.2.23)$$

This window function has a magnitude response

$$|W(\omega)| = \frac{|\sin(\omega M/2)|}{|\sin(\omega/2)|} \quad -\pi \leq \omega \leq \pi \quad (8.2.24)$$

and a piecewise linear phase

$$\Theta(\omega) = \begin{cases} -\omega \left(\frac{M-1}{2} \right), & \text{when } \sin(\omega M/2) \geq 0 \\ -\omega \left(\frac{M-1}{2} \right) + \pi, & \text{when } \sin(\omega M/2) < 0 \end{cases} \quad (8.2.25)$$

The magnitude response of the window function is illustrated in Fig. 8.4 for $M = 31$ and 61. The width of the main lobe [width is measured to the first zero of $W(\omega)$]

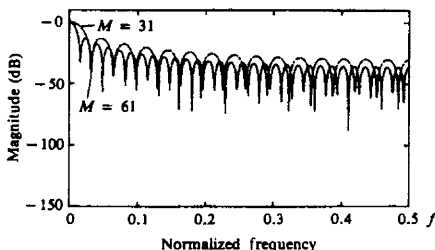


Figure 8.4 Frequency response for rectangular window of lengths (a) $M = 31$, (b) $M = 61$.

is $4\pi/M$. Hence, as M increases, the main lobe becomes narrower. However, the sidelobes of $|W(\omega)|$ are relatively high and remain unaffected by an increase in M . In fact, even though the width of each sidelobe decreases with an increase in M , the height of each sidelobe increases with an increase in M in such a manner that the area under each sidelobe remains invariant to changes in M . This characteristic behavior is not evident from observation of Fig. 8.4 because $W(\omega)$ has been normalized by M such that the normalized peak values of the sidelobes remain invariant to an increase in M .

The characteristics of the rectangular window play a significant role in determining the resulting frequency response of the FIR filter obtained by truncating $h_d(n)$ to length M . Specifically, the convolution of $H_d(\omega)$ with $W(\omega)$ has the effect of smoothing $H_d(\omega)$. As M is increased, $W(\omega)$ becomes narrower, and the smoothing provided by $W(\omega)$ is reduced. On the other hand, the large sidelobes of $W(\omega)$ result in some undesirable ringing effects in the FIR filter frequency response $H(\omega)$, and also in relatively larger sidelobes in $H(\omega)$. These undesirable effects are best alleviated by the use of windows that do not contain abrupt discontinuities in their time-domain characteristics, and have correspondingly low sidelobes in their frequency-domain characteristics.

Table 8.1 lists several window functions that possess desirable frequency response characteristics. Figure 8.5 illustrates the time-domain characteristics of the windows. The frequency response characteristics of the Hanning, Hamming, and Blackman windows are illustrated in Figs. 8.6 through 8.8. All of these window functions have significantly lower sidelobes compared with the rectangular window. However, for the same value of M , the width of the main lobe is also wider for these windows compared to the rectangular window. Consequently, these window functions provide more smoothing through the convolution operation in the frequency domain, and as a result, the transition region in the FIR filter response is wider. To reduce the width of this transition region, we can simply increase the length of the window which results in a larger filter. Table 8.2 summarizes these important frequency-domain features of the various window functions.

The window technique is best described in terms of a specific example. Suppose that we want to design a symmetric lowpass linear-phase FIR filter having a

TABLE 8.1 WINDOW FUNCTIONS FOR FIR FILTER DESIGN

Name of window	Time-domain sequence, $h(n), 0 \leq n \leq M-1$
Bartlett (triangular)	$1 - \frac{2 \left n - \frac{M-1}{2} \right }{M-1}$
Blackman	$0.42 - 0.5 \cos \frac{2\pi n}{M-1} + 0.08 \cos \frac{4\pi n}{M-1}$
Hamming	$0.54 - 0.46 \cos \frac{2\pi n}{M-1}$
Hanning	$\frac{1}{2} \left(1 - \cos \frac{2\pi n}{M-1} \right)$
Kaiser	$\frac{I_0 \left[\alpha \sqrt{\left(\frac{M-1}{2} \right)^2 - \left(n - \frac{M-1}{2} \right)^2} \right]}{I_0 \left[\alpha \left(\frac{M-1}{2} \right) \right]}$
Lanczos	$\left\{ \frac{\sin \left[2\pi \left(n - \frac{M-1}{2} \right) / (M-1) \right]}{2\pi \left(n - \frac{M-1}{2} \right) / \left(\frac{M-1}{2} \right)} \right\}^L \quad L > 0$
Tukey	$\frac{1}{2} \left[1 + \cos \left(\frac{n - (1+\alpha)(M-1)/2}{(1-\alpha)(M-1)/2} \pi \right) \right]$ $\alpha(M-1)/2 \leq \left n - \frac{M-1}{2} \right \leq \frac{M-1}{2}$

desired frequency response

$$H_d(\omega) = \begin{cases} 1e^{-j\omega(M-1)/2}, & 0 \leq |\omega| \leq \omega_c \\ 0, & \text{otherwise} \end{cases} \quad (8.2.26)$$

A delay of $(M-1)/2$ units is incorporated into $H_d(\omega)$ in anticipation of forcing the filter to be of length M . The corresponding unit sample response, obtained by evaluating the integral in (8.2.18), is

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n - \frac{M-1}{2})} d\omega$$

$$= \frac{\sin \omega_c \left(n - \frac{M-1}{2} \right)}{\pi \left(n - \frac{M-1}{2} \right)} \quad n \neq \frac{M-1}{2} \quad (8.2.27)$$

Clearly, $h_d(n)$ is noncausal and infinite in duration.

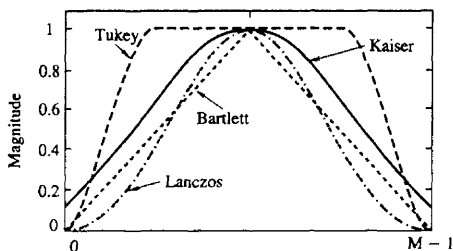
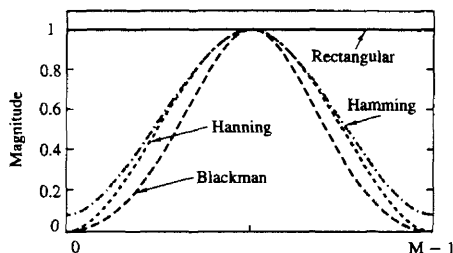


Figure 8.5 Shapes of several window functions.

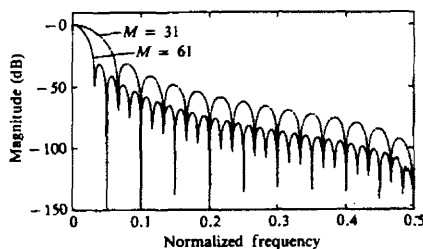


Figure 8.6 Frequency responses of Hanning window for (a) $M = 31$ and (b) $M = 61$.

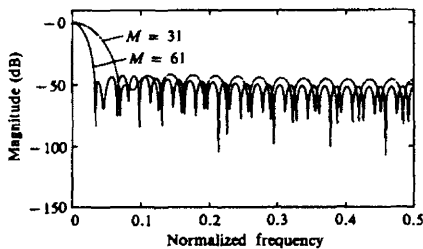


Figure 8.7 Frequency responses for Hamming window for (a) $M = 31$ and (b) $M = 61$.

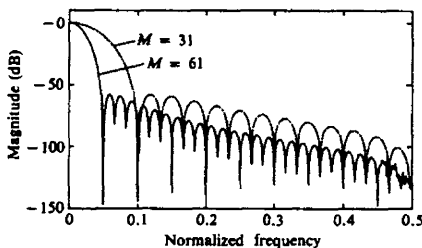


Figure 8.8 Frequency responses for Blackman window for (a) $M = 31$ and (b) $M = 61$.

TABLE 8.2 IMPORTANT FREQUENCY-DOMAIN CHARACTERISTICS OF SOME WINDOW FUNCTIONS

Type of window	Approximate transition width of main lobe	Peak sidelobe (dB)
Rectangular	$4\pi/M$	-13
Bartlett	$8\pi/M$	-27
Hanning	$8\pi/M$	-32
Hamming	$8\pi/M$	-43
Blackman	$12\pi/M$	-58

If we multiply $h_d(n)$ by the rectangular window sequence in (8.2.19), we obtain an FIR filter of length M having the unit sample response

$$h(n) = \frac{\sin \omega_c \left(n - \frac{M-1}{2} \right)}{\pi \left(n - \frac{M-1}{2} \right)} \quad 0 \leq n \leq M-1 \quad n \neq \frac{M-1}{2} \quad (8.2.28)$$

If M is selected to be odd, the value of $h(n)$ at $n = (M-1)/2$ is

$$h \left(\frac{M-1}{2} \right) = \frac{\omega_c}{\pi} \quad (8.2.29)$$

The magnitude of the frequency response $H(\omega)$ of this filter is illustrated in Fig. 8.9 for $M = 61$ and $M = 101$. We observe that relatively large oscillations or ripples occur near the band edge of the filter. The oscillations increase in frequency as M increases, but they do not diminish in amplitude. As indicated previously, these large oscillations are the direct result of the large sidelobes existing in the frequency characteristic $W(\omega)$ of the rectangular window. As this window function is convolved with the desired frequency response characteristic $H_d(\omega)$, the oscillations occur as the large constant area sidelobes of $W(\omega)$ move across the discontinuity that exists in $H_d(\omega)$. Since (8.2.17) is basically a Fourier series representation of $H_d(\omega)$, the multiplication of $h_d(n)$ with a rectangular window is identical to truncating the Fourier series representation of the desired filter char-

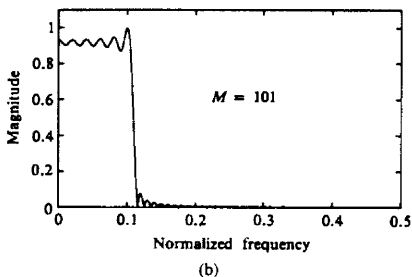
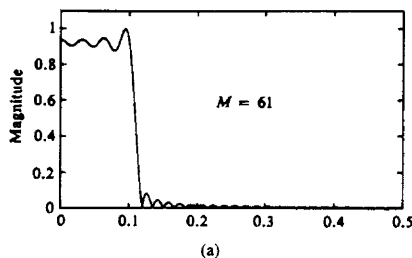


Figure 8.9 Lowpass filter designed with a rectangular window (a) $M = 61$ and (b) $M = 101$.

acteristic $H_d(\omega)$. The truncation of the Fourier series is known to introduce ripples in the frequency response characteristic $H(\omega)$ due to the nonuniform convergence of the Fourier series at a discontinuity. The oscillatory behavior near the band edge of the filter is called the *Gibbs phenomenon*.

To alleviate the presence of large oscillations in both the passband and the stopband, we should use a window function that contains a taper and decays toward zero gradually, instead of abruptly, as it occurs in a rectangular window. Figures 8.10 through 8.13, illustrate the frequency response of the resulting filter when some of the window functions listed in Table 8.1 are used to taper $h_d(n)$. As illustrated in Figs. 8.10 through 8.13, the window functions do indeed eliminate the ringing effects at the band edge and do result in lower sidelobes at the expense of an increase in the width of the transition band of the filter.

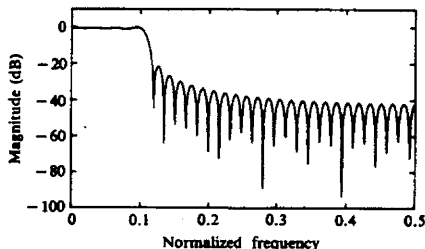


Figure 8.10 Lowpass FIR filter designed with rectangular window ($M = 61$).

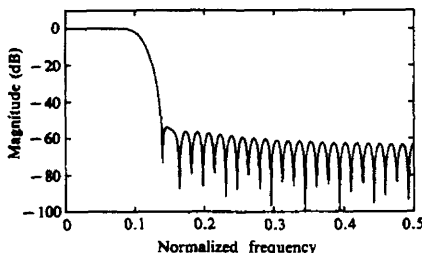


Figure 8.11 Lowpass FIR filter designed with Hamming window ($M = 61$).

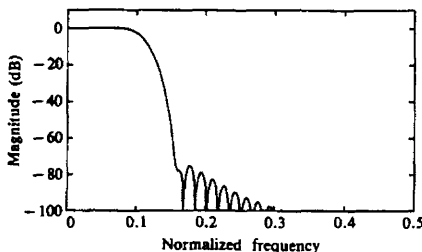


Figure 8.12 Lowpass FIR filter designed with Blackman window ($M = 61$).

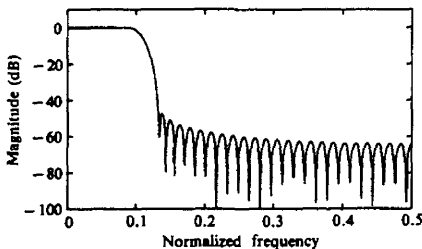


Figure 8.13 Lowpass FIR filter designed with $\alpha = 4$ Kaiser window ($M = 61$).

8.2.3 Design of Linear-Phase FIR Filters by the Frequency-Sampling Method

In the frequency sampling method for FIR filter design, we specify the desired frequency response $H_d(\omega)$ at a set of equally spaced frequencies, namely

$$\begin{aligned} \omega_k &= \frac{2\pi}{M}(k + \alpha) & k = 0, 1, \dots, \frac{M-1}{2} & \quad M \text{ odd} \\ & & k = 0, 1, \dots, \frac{M}{2} - 1 & \quad M \text{ even} \\ \alpha &= 0 \quad \text{or} \quad \frac{1}{2} \end{aligned} \quad (8.2.30)$$

and solve for the unit sample response $h(n)$ of the FIR filter from these equally

spaced frequency specifications. To reduce sidelobes, it is desirable to optimize the frequency specification in the transition band of the filter. This optimization can be accomplished numerically on a digital computer by means of linear programming techniques as shown by Rabiner et al. (1970).

In this section we exploit a basic symmetry property of the sampled frequency response function to simplify the computations. Let us begin with the desired frequency response of the FIR filter, which is [for simplicity, we drop the subscript in $H_d(\omega)$],

$$H(\omega) = \sum_{n=0}^{M-1} h(n)e^{-j\omega n} \quad (8.2.31)$$

Suppose that we specify the frequency response of the filter at the frequencies given by (8.2.30). Then from (8.2.31) we obtain

$$H(k + \alpha) \equiv H\left(\frac{2\pi}{M}(k + \alpha)\right)$$

$$H(k + \alpha) \equiv \sum_{n=0}^{M-1} h(n)e^{-j2\pi(k+\alpha)n/M} \quad k = 0, 1, \dots, M-1 \quad (8.2.32)$$

It is a simple matter to invert (8.2.32) and express $h(n)$ in terms of $H(k + \alpha)$. If we multiply both sides of (8.2.32) by the exponential, $\exp(j2\pi km/M)$, $m = 0, 1, \dots, M-1$, and sum over $k = 0, 1, \dots, M-1$, the right-hand side of (8.2.32) reduces to $Mh(m)\exp(-j2\pi\alpha m/M)$. Thus we obtain

$$h(n) = \frac{1}{M} \sum_{k=0}^{M-1} H(k + \alpha)e^{j2\pi(k+\alpha)n/M} \quad n = 0, 1, \dots, M-1 \quad (8.2.33)$$

The relationship in (8.2.33) allows us to compute the values of the unit sample response $h(n)$ from the specification of the frequency samples $H(k + \alpha)$, $k = 0, 1, \dots, M-1$. Note that when $\alpha = 0$, (8.2.32) reduces to the discrete Fourier transform (DFT) of the sequence $\{h(n)\}$ and (8.2.33) reduces to the inverse DFT (IDFT).

Since $\{h(n)\}$ is real, we can easily show that the frequency samples $\{H(k + \alpha)\}$ satisfy the symmetry condition

$$H(k + \alpha) = H^*(M - k - \alpha) \quad (8.2.34)$$

This symmetry condition, along with the symmetry conditions for $\{h(n)\}$, can be used to reduce the frequency specifications from M points to $(M+1)/2$ points for M odd and $M/2$ points for M even. Thus the linear equations for determining $\{h(n)\}$ from $\{H(k + \alpha)\}$ are considerably simplified.

In particular, if (8.2.11) is sampled at the frequencies $\omega_k = 2\pi(k + \alpha)/M$, $k = 0, 1, \dots, M-1$, we obtain

$$H(k + \alpha) = H_r\left(\frac{2\pi}{M}(k + \alpha)\right)e^{j[\delta\pi/2 - 2\pi(k+\alpha)(M-1)/2M]} \quad (8.2.35)$$

where $\beta = 0$ when $\{h(n)\}$ is symmetric and $\beta = 1$ when $\{h(n)\}$ is antisymmetric. A simplification occurs by defining a set of real frequency samples $\{G(k+m)\}$

$$G(k+\alpha) = (-1)^k H_r \left(\frac{2\pi}{M}(k+\alpha) \right) \quad k = 0, 1, \dots, M-1 \quad (8.2.36)$$

We use (8.2.36) in (8.2.35) to eliminate $H_r(\omega_k)$. Thus we obtain

$$H(k+\alpha) = G(k+\alpha) e^{j\pi k} e^{j[\beta\pi/2 - 2\pi(k+\alpha)(M-1)/2M]} \quad (8.2.37)$$

Now the symmetry condition for $H(k+\alpha)$ given in (8.2.34) translates into a corresponding symmetry condition for $G(k+\alpha)$, which can be exploited by substituting into (8.2.33), to simplify the expressions for the FIR filter impulse response $\{h(n)\}$ for the four cases $\alpha = 0$, $\alpha = \frac{1}{2}$, $\beta = 0$, and $\beta = 1$. The results are summarized in Table 8.3. The detailed derivations are left as exercises for the reader.

Although the frequency sampling method provides us with another means for designing linear-phase FIR filters, its major advantage lies in the efficient frequency sampling structure, which is obtained when most of the frequency samples are zero, as demonstrated in Section 7.2.3.

The following examples illustrate the design of linear-phase FIR filters based on the frequency sampling method. The optimum values for the samples in the transition band are obtained from the tables in Appendix C which are taken from the paper by Rabiner et al. (1970).

Example 8.2.1

Determine the coefficients of a linear-phase FIR filter of length $M = 15$ which has a symmetric unit sample response and a frequency response that satisfies the conditions

$$H_r \left(\frac{2\pi k}{15} \right) = \begin{cases} 1, & k = 0, 1, 2, 3 \\ 0.4, & k = 4 \\ 0, & k = 5, 6, 7 \end{cases}$$

Solution Since $h(n)$ is symmetric and the frequencies are selected to correspond to the case $\alpha = 0$, we use the corresponding formula in Table 8.3 to evaluate $h(n)$. In this case

$$G(k) = (-1)^k H_r \left(\frac{2\pi k}{15} \right) \quad k = 0, 1, \dots, 7$$

The result of this computation is

$$\begin{aligned} h(0) &= h(14) = -0.014112893 \\ h(1) &= h(13) = -0.001945309 \\ h(2) &= h(12) = 0.040000004 \\ h(3) &= h(11) = 0.01223454 \\ h(4) &= h(10) = -0.09138802 \\ h(5) &= h(9) = -0.01808986 \\ h(6) &= h(8) = 0.3133176 \\ h(7) &= 0.52 \end{aligned}$$

TABLE 8.3 UNIT SAMPLE RESPONSE: $h(n) = \pm h(M-1-n)$

Symmetric	
	$H(k) = G(k)e^{j\pi k/M} \quad k = 0, 1, \dots, M-1$
$\alpha = 0$	$G(k) = (-1)^k H_r\left(\frac{2\pi k}{M}\right) \quad G(k) = -G(M-k)$ $h(n) = \frac{1}{M} \left\{ G(0) + 2 \sum_{k=1}^U G(k) \cos \frac{2\pi k}{M} \left(n + \frac{1}{2}\right) \right\}$ $U = \begin{cases} \frac{M-1}{2}, & M \text{ odd} \\ \frac{M}{2} - 1, & M \text{ even} \end{cases}$ $H\left(k + \frac{1}{2}\right) = G\left(k + \frac{1}{2}\right) e^{-j\pi/2} e^{j\pi(2k+1)/2M}$
$\alpha = \frac{1}{2}$	$G\left(k + \frac{1}{2}\right) = (-1)^k H_r\left[\frac{2\pi}{M}\left(k + \frac{1}{2}\right)\right]$ $G\left(k + \frac{1}{2}\right) = G\left(M - k - \frac{1}{2}\right)$ $h(n) = \frac{2}{M} \sum_{k=0}^U G\left(k + \frac{1}{2}\right) \sin \frac{2\pi}{M}\left(k + \frac{1}{2}\right)\left(n + \frac{1}{2}\right)$
Antisymmetric	
	$H(k) = G(k)e^{j\pi/2} e^{j\pi k/M} \quad k = 0, 1, \dots, M-1$
$\alpha = 0$	$G(k) = (-1)^k H_r\left(\frac{2\pi k}{M}\right) \quad G(k) = G(M-k)$ $h(n) = -\frac{2}{M} \sum_{k=1}^{(M-1)/2} G(k) \sin \frac{2\pi k}{M} \left(n + \frac{1}{2}\right) \quad M \text{ odd}$ $h(n) = \frac{1}{M} \left\{ (-1)^{n+1} G(M/2) - 2 \sum_{k=1}^{(M/2)-1} G(k) \sin \frac{2\pi k}{M} \left(n + \frac{1}{2}\right) \right\} \quad M \text{ even}$ $H\left(k + \frac{1}{2}\right) = G\left(k + \frac{1}{2}\right) e^{j\pi(2k+1)/2M}$
$\alpha = \frac{1}{2}$	$G\left(k + \frac{1}{2}\right) = (-1)^k H_r\left[\frac{2\pi}{M}\left(k + \frac{1}{2}\right)\right]$ $G\left(k + \frac{1}{2}\right) = -G\left(M - k - \frac{1}{2}\right); G(M/2) = 0 \text{ for } M \text{ odd}$ $h(n) = \frac{2}{M} \sum_{k=0}^U G\left(k + \frac{1}{2}\right) \cos \frac{2\pi}{M}\left(k + \frac{1}{2}\right)\left(n + \frac{1}{2}\right)$ $V = \begin{cases} \frac{M-3}{2}, & M \text{ odd} \\ \frac{M}{2} - 1, & M \text{ even} \end{cases}$

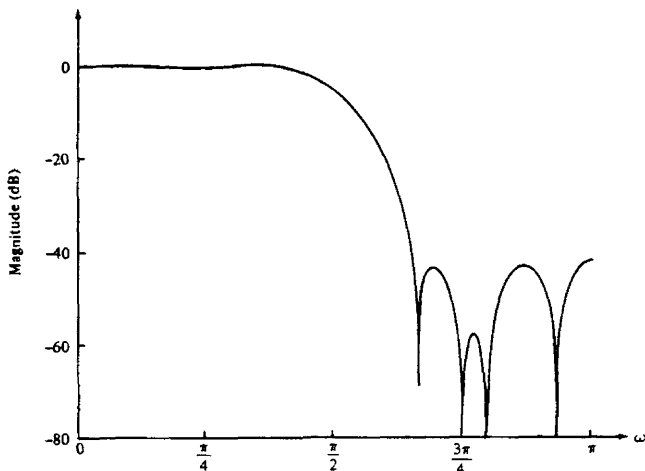


Figure 8.14 Frequency response of linear-phase FIR filter in Example 8.2.1.

The frequency response characteristic of this filter is shown in Fig. 8.14. We should emphasize that $H_r(\omega)$ is exactly equal to the values given by the specifications above at $\omega_k = 2\pi k/15$.

Example 8.2.2

Determine the coefficients of a linear-phase FIR filter of length $M = 32$ which has a symmetric unit sample response and a frequency response that satisfies the condition

$$H_r\left(\frac{2\pi(k+\alpha)}{32}\right) = \begin{cases} 1, & k = 0, 1, 2, 3, 4, 5 \\ T_1, & k = 6 \\ 0, & k = 7, 8, \dots, 15 \end{cases}$$

where $T_1 = 0.3789795$ for $\alpha = 0$, and $T_1 = 0.3570496$ for $\alpha = \frac{1}{2}$. These values of T_1 were obtained from the tables of optimum transition parameters given in Appendix C.

Solution The appropriate equations for this computation are given in Table 8.3 for $\alpha = 0$ and $\alpha = \frac{1}{2}$. These computations yield the unit sample responses shown in Table 8.4. The corresponding frequency response characteristics are illustrated in Figs. 8.15 and 8.16, respectively. Note that the bandwidth of the filter for $\alpha = \frac{1}{2}$ is wider than that for $\alpha = 0$.

The optimization of the frequency samples in the transition region of the frequency response can be explained by evaluating the system function $H(z)$, given by (7.2.12), on the unit circle and using the relationship in (8.2.37) to express $H(\omega)$

TABLE 8.4

$M = 32$	$M = 32$
$\text{ALPHA} = 0.$	$\text{ALPHA} = 0.5$
$T1 = 0.3789795\text{E}+00$	$T1 = 0.3570496\text{E}+00$
$h(0) = -0.7141978\text{E}-02$	$h(0) = -0.4089120\text{E}-02$
$h(1) = -0.3070801\text{E}-02$	$h(1) = -0.9973779\text{E}-02$
$h(2) = 0.5891327\text{E}-02$	$h(2) = -0.7379891\text{E}-02$
$h(3) = 0.1349923\text{E}-01$	$h(3) = 0.5949799\text{E}-02$
$h(4) = 0.8087033\text{E}-02$	$h(4) = 0.1727056\text{E}-01$
$h(5) = -0.1107258\text{E}-01$	$h(5) = 0.7878412\text{E}-02$
$h(6) = -0.2420687\text{E}-01$	$h(6) = -0.1798590\text{E}-01$
$h(7) = -0.9446550\text{E}-02$	$h(7) = -0.2670584\text{E}-01$
$h(8) = 0.2544464\text{E}-01$	$h(8) = 0.3778549\text{E}-02$
$h(9) = 0.3985050\text{E}-01$	$h(9) = 0.4191022\text{E}-01$
$h(10) = 0.2753036\text{E}-02$	$h(10) = 0.2839344\text{E}-01$
$h(11) = -0.5913959\text{E}-01$	$h(11) = -0.4163144\text{E}-01$
$h(12) = -0.6841660\text{E}-01$	$h(12) = -0.8254962\text{E}-01$
$h(13) = 0.3175741\text{E}-01$	$h(13) = 0.2802212\text{E}-02$
$h(14) = 0.2080981\text{E}+00$	$h(14) = 0.2013655\text{E}+00$
$h(15) = 0.3471138\text{E}+00$	$h(15) = 0.3717532\text{E}+00$

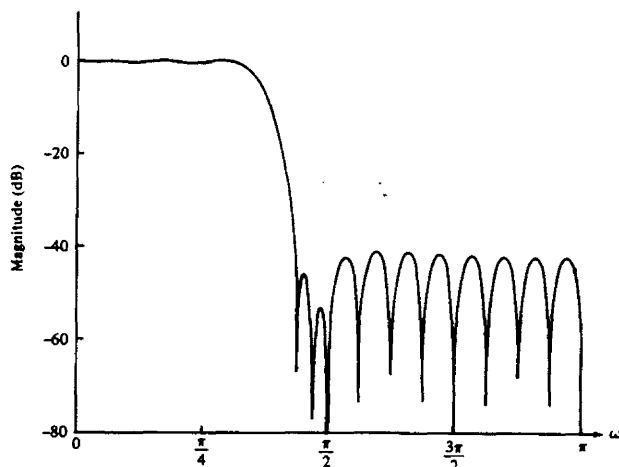


Figure 8.15 Frequency response of linear-phase FIR filter in Example 8.2.2 ($M = 32$ and $\alpha = 0$).

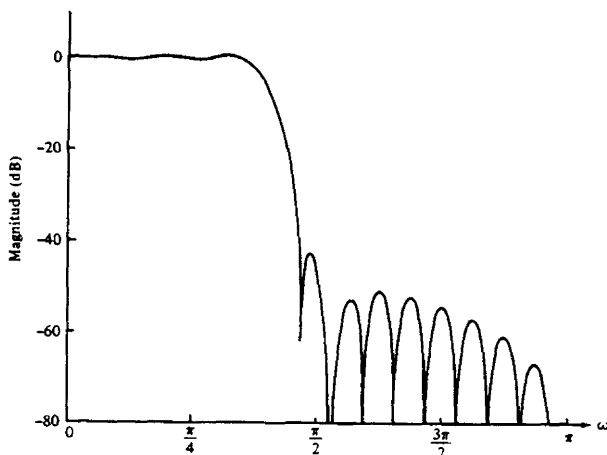


Figure 8.16 Frequency response of linear-phase FIR filter in Example 8.2.2 ($M = 32$ and $\alpha = \frac{1}{2}$).

in terms of $G(k + \alpha)$. Thus for the symmetric filter we obtain

$$H(\omega) = \left\{ \frac{\sin\left(\frac{\omega M}{2} - \pi\alpha\right)}{M} \sum_{k=0}^{M-1} \frac{G(k + \alpha)}{\sin\left[\frac{\omega}{2} - \frac{\pi}{M}(k + \alpha)\right]} \right\} e^{-j\omega(M-1)/2} \quad (8.2.38)$$

where

$$G(k + \alpha) = \begin{cases} -G(M - k), & \alpha = 0 \\ G(M - k - \frac{1}{2}), & \alpha = \frac{1}{2} \end{cases} \quad (8.2.39)$$

Similarly, for the antisymmetric linear-phase FIR filter we obtain

$$H(\omega) = \left\{ \frac{\sin\left(\frac{\omega M}{2} - \pi\alpha\right)}{M} \sum_{k=0}^{M-1} \frac{G(k + \alpha)}{\sin\left[\frac{\omega}{2} - \frac{\pi}{M}(k + \alpha)\right]} \right\} e^{-j\omega(M-1)/2} e^{j\pi/2} \quad (8.2.40)$$

where

$$G(k + \alpha) = \begin{cases} G(M - k), & \alpha = 0 \\ -G(M - k - \frac{1}{2}), & \alpha = \frac{1}{2} \end{cases} \quad (8.2.41)$$

With these expressions for the frequency response $H(\omega)$ given in terms of the desired frequency samples $\{G(k + \alpha)\}$, we can easily explain the method for selecting the parameters $\{G(k + \alpha)\}$ in the transition band which result in minimizing the peak sidelobe in the stopband. In brief, the values of $G(k + \alpha)$ in the passband are

set to $(-1)^k$ and those in the stopband are set to zero. For any choice of $G(k + \alpha)$ in the transition band, the value of $H(\omega)$ is computed at a dense set of frequencies (e.g., at $\omega_n = 2\pi n/K$, $n = 0, 1, \dots, K - 1$, where, for example, $K = 10M$). The value of the maximum sidelobe is determined, and the values of the parameters $\{G(k + \alpha)\}$ in the transition band are changed in a direction of steepest descent, which, in effect, reduces the maximum sidelobe. The computation of $H(\omega)$ is now repeated with the new choice of $\{G(k + \alpha)\}$. The maximum sidelobe of $H(\omega)$ is again determined and the values of the parameters $\{G(k + \alpha)\}$ in the transition band are adjusted in a direction of steepest descent that, in turn, reduces the sidelobe. This interactive process is performed until it converges to the optimum choice of the parameters $\{G(k + \alpha)\}$ in the transition band.

There is a potential problem in the frequency-sampling realization of the FIR linear-phase filter. The frequency sampling realization of the FIR filter introduces poles and zeros at equally spaced points on the unit circle. In the ideal situation, the zeros cancel the poles and, consequently, the actual zeros of $H(z)$ are determined by the selection of the frequency samples $\{H(k + \alpha)\}$. In a practical implementation of the frequency-sampling realization, however, quantization effects preclude a perfect cancellation of the poles and zeros. In fact, the location of poles on the unit circle provide no damping of the round-off noise that is introduced in the computations. As a result, such noise tends to increase with time and, ultimately, may destroy the normal operation of the filter.

To mitigate this problem, we can move both the poles and zeros from the unit circle to a circle just inside the unit circle, say at radius $r = 1 - \epsilon$, where ϵ is a very small number. Thus the system function of the linear-phase FIR filter becomes

$$H(z) = \frac{1 - r^M z^{-M} e^{j2\pi\alpha}}{M} \sum_{k=0}^{M-1} \frac{H(k + \alpha)}{1 - r e^{j2\omega\pi(k+\alpha)/M} z^{-1}} \quad (8.2.42)$$

The corresponding two-pole filter realization given in Section 7.2.3 can be modified accordingly. The damping provided by selecting $r < 1$ ensures that roundoff noise will be bounded and thus instability is avoided.

8.2.4 Design of Optimum Equiripple Linear-Phase FIR Filters

The window method and the frequency-sampling method are relatively simple techniques for designing linear-phase FIR filters. However, they also possess some minor disadvantages, described in Section 8.2.6, which may render them undesirable for some applications. A major problem is the lack of precise control of the critical frequencies such as ω_p and ω_s .

The filter design method described in this section is formulated as a Chebyshev approximation problem. It is viewed as an optimum design criterion in the sense that the weighted approximation error between the desired frequency response and the actual frequency response is spread evenly across the passband

and evenly across the stopband of the filter minimizing the maximum error. The resulting filter designs have ripples in both the passband and the stopband.

To describe the design procedure, let us consider the design of a lowpass filter with passband edge frequency ω_p and stopband edge frequency ω_s . From the general specifications given in Fig. 8.2, in the passband, the filter frequency response satisfies the condition

$$1 - \delta_1 \leq H_r(\omega) \leq 1 + \delta_1 \quad |\omega| \leq \omega_p \quad (8.2.43)$$

Similarly, in the stopband, the filter frequency response is specified to fall between the limits $\pm\delta_2$, that is,

$$-\delta_2 \leq H_r(\omega) \leq \delta_2 \quad |\omega| > \omega_s \quad (8.2.44)$$

Thus δ_1 represents the ripple in the passband and δ_2 represents the attenuation or ripple in the stopband. The remaining filter parameter is M , the filter length or the number of filter coefficients.

Let us focus on the four different cases that result in a linear-phase FIR filter. These cases were treated in Section 8.2.2 and are summarized below.

Case 1: Symmetric unit sample response $h(n) = h(M-1-n)$ and M Odd. In this case, the real-valued frequency response characteristic $H_r(\omega)$ is

$$H_r(\omega) = h\left(\frac{M-1}{2}\right) + 2 \sum_{n=0}^{(M-3)/2} h(n) \cos \omega \left(\frac{M-1}{2} - n\right) \quad (8.2.45)$$

If we let $k = (M-1)/2 - n$ and define a new set of filter parameters $\{a(k)\}$ as

$$a(k) = \begin{cases} h\left(\frac{M-1}{2}\right), & k = 0 \\ 2h\left(\frac{M-1}{2} - k\right), & k = 1, 2, \dots, \frac{M-1}{2} \end{cases} \quad (8.2.46)$$

then (8.2.45) reduces to the compact form

$$H_r(\omega) = \sum_{k=0}^{(M-1)/2} a(k) \cos \omega k \quad (8.2.47)$$

Case 2: Symmetric unit sample response $h(n) = h(M-1-n)$ and M Even. In this case, $H_r(\omega)$ is expressed as

$$H_r(\omega) = 2 \sum_{n=0}^{(M/2)-1} h(n) \cos \omega \left(\frac{M-1}{2} - n\right) \quad (8.2.48)$$

Again, we change the summation index from n to $k = M/2 - n$ and define a new set of filter parameters $\{b(k)\}$ as

$$b(k) = 2h\left(\frac{M}{2} - k\right), \quad k = 1, 2, \dots, M/2 \quad (8.2.49)$$

With these substitutions (8.2.48) becomes

$$H_r(\omega) = \sum_{k=1}^{M/2} b(k) \cos \omega \left(k - \frac{1}{2} \right) \quad (8.2.50)$$

In carrying out the optimization, it is convenient to rearrange (8.2.50) further into the form

$$H_r(\omega) = \cos \frac{\omega}{2} \sum_{k=0}^{(M/2)-1} \tilde{b}(k) \cos \omega k \quad (8.2.51)$$

where the coefficients $\{\tilde{b}(k)\}$ are linearly related to the coefficients $\{b(k)\}$. In fact, it can be shown that the relationship is

$$\begin{aligned} \tilde{b}(0) &= \frac{1}{2}b(1) \\ \tilde{b}(k) &= 2b(k) - \tilde{b}(k-1) \quad k = 1, 2, 3, \dots, \frac{M}{2} - 2 \\ \tilde{b}\left(\frac{M}{2} - 1\right) &= 2b\left(\frac{M}{2}\right) \end{aligned} \quad (8.2.52)$$

Case 3: Antisymmetric unit sample response $h(n) = -h(M-1-n)$ and M Odd. The real-valued frequency response characteristic $H_r(\omega)$ for this case is

$$H_r(\omega) = 2 \sum_{n=0}^{(M-3)/2} h(n) \sin \omega \left(\frac{M-1}{2} - n \right) \quad (8.2.53)$$

If we change the summation in (8.2.53) from n to $k = (M-1)/2 - n$ and define a new set of filter parameters $\{c(k)\}$ as

$$c(k) = 2h\left(\frac{M-1}{2} - k\right) \quad k = 1, 2, \dots, (M-1)/2 \quad (8.2.54)$$

then (8.2.53) becomes

$$H_r(\omega) = \sum_{k=1}^{(M-1)/2} c(k) \sin \omega k \quad (8.2.55)$$

As in the previous case, it is convenient to rearrange (8.2.55) into the form

$$H_r(\omega) = \sin \omega \sum_{k=0}^{(M-3)/2} \tilde{c}(k) \cos \omega k \quad (8.2.56)$$

where the coefficients $\{\tilde{c}(k)\}$ are linearly related to the parameters $\{c(k)\}$. This desired relationship can be derived from (8.2.55) and (8.2.56) and is simply

given as

$$\begin{aligned}
 \tilde{c}\left(\frac{M-3}{2}\right) &= c\left(\frac{M-1}{2}\right) \\
 \tilde{c}\left(\frac{M-5}{2}\right) &= 2c\left(\frac{M-3}{2}\right) \\
 &\vdots \\
 \tilde{c}(k-1) - \tilde{c}(k+1) &= 2c(k) \quad 2 \leq k \leq \frac{M-5}{2} \\
 \tilde{c}(0) + \frac{1}{2}\tilde{c}(2) &= c(1)
 \end{aligned} \tag{8.2.57}$$

Case 4: Antisymmetric unit sample response $h(n) = -h(M-1-n)$ and M Even. In this case, the real-valued frequency response characteristic $H_r(\omega)$ is

$$H_r(\omega) = 2 \sum_{n=0}^{(M/2)-1} h(n) \sin \omega \left(\frac{M-1}{2} - n \right) \tag{8.2.58}$$

A change in the summation index from n to $k = M/2 - n$ combined with a definition of a new set of filter coefficients $\{d(k)\}$, related to $\{h(n)\}$ according to

$$d(k) = 2h\left(\frac{M}{2} - k\right) \quad k = 1, 2, \dots, \frac{M}{2} \tag{8.2.59}$$

results in the expression

$$H_r(\omega) = \sum_{k=1}^{M/2} d(k) \sin \omega \left(k - \frac{1}{2} \right) \tag{8.2.60}$$

As in the previous two cases, we find it convenient to rearrange (8.2.60) into the form

$$H_r(\omega) = \sin \frac{\omega}{2} \sum_{k=0}^{(M/2)-1} \tilde{d}(k) \cos \omega k \tag{8.2.61}$$

where the new filter parameters $\{\tilde{d}(k)\}$ are related to $\{d(k)\}$ as follows:

$$\begin{aligned}
 \tilde{d}\left(\frac{M}{2} - 1\right) &= 2d\left(\frac{M}{2}\right) \\
 \tilde{d}(k-1) - \tilde{d}(k) &= 2d(k) \quad 2 \leq k \leq \frac{M}{2} - 1 \\
 \tilde{d}(0) - \frac{1}{2}\tilde{d}(1) &= d(1)
 \end{aligned} \tag{8.2.62}$$

The expressions for $H_r(\omega)$ in these four cases are summarized in Table 8.5. We note that the rearrangements that we made in cases 2, 3, and 4 have allowed

TABLE 8.5 REAL-VALUED FREQUENCY RESPONSE FUNCTIONS FOR LINEAR-PHASE FIR FILTERS

Filter type	$Q(\omega)$	$P(\omega)$
$h(n) = h(M-1-n)$ M odd (case 1)	1	$\sum_{k=0}^{(M-1)/2} a(k) \cos \omega k$
$h(n) = h(M-1-n)$ M even (case 2)	$\cos \frac{\omega}{2}$	$\sum_{k=0}^{(M/2)-1} \tilde{b}(k) \cos \omega k$
$h(n) = -h(M-1-n)$ M odd (case 3)	$\sin \omega$	$\sum_{k=0}^{(M-3)/2} \tilde{c}(k) \cos \omega k$
$h(n) = -h(M-1-n)$ M even (case 4)	$\sin \frac{\omega}{2}$	$\sum_{k=0}^{(M/2)-1} \tilde{d}(k) \cos \omega k$

us to express $H_r(\omega)$ as

$$H_r(\omega) = Q(\omega)P(\omega) \quad (8.2.63)$$

where

$$Q(\omega) = \begin{cases} 1 & \text{case 1} \\ \cos \frac{\omega}{2} & \text{case 2} \\ \sin \omega & \text{case 3} \\ \sin \frac{\omega}{2} & \text{case 4} \end{cases} \quad (8.2.64)$$

and $P(\omega)$ has the common form

$$P(\omega) = \sum_{k=0}^L \alpha(k) \cos \omega k \quad (8.2.65)$$

with $\{\alpha(k)\}$ representing the parameters of the filter, which are linearly related to the unit sample response $h(n)$ of the FIR filter. The upper limit L in the sum is $L = (M-1)/2$ for Case 1, $L = (M-3)/2$ for Case 3, and $L = M/2 - 1$ for Case 2 and Case 4.

In addition to the common framework given above for the representation of $H_r(\omega)$, we also define the real-valued desired frequency response $H_{dr}(\omega)$ and the weighting function $W(\omega)$ on the approximation error. The real-valued desired frequency response $H_{dr}(\omega)$ is simply defined to be unity in the passband and zero in the stopband. For example, Fig. 8.17 illustrates several different types of characteristics for $H_{dr}(\omega)$. The weighting function on the approximation error allows us to choose the relative size of the errors in the different frequency bands (i.e., in the passband and in the stopband). In particular, it is convenient to normalize $W(\omega)$ to unity in the stopband and set $W(\omega) = \delta_2/\delta_1$ in the passband,

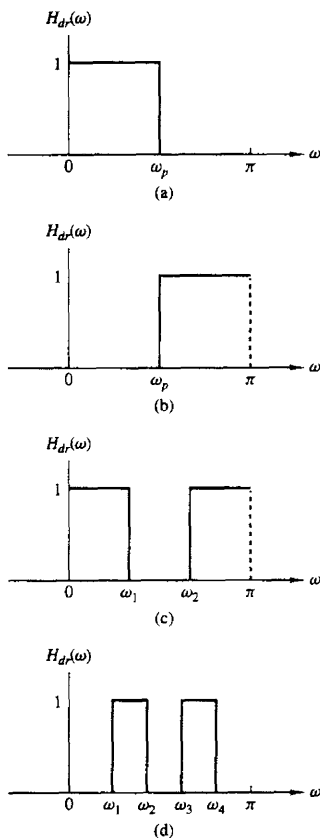


Figure 8.17 Desired frequency response characteristics for different types of filters.

that is,

$$W(\omega) = \begin{cases} \delta_2/\delta_1, & \omega \text{ in the passband} \\ 1, & \omega \text{ in the stopband} \end{cases} \quad (8.2.66)$$

Then we simply select $W(\omega)$ in the passband to reflect our emphasis on the relative size of the ripple in the stopband to the ripple in the passband.

With the specification of $H_d(\omega)$ and $W(\omega)$, we can now define the weighted approximation error as

$$\begin{aligned} E(\omega) &= W(\omega)[H_d(\omega) - H_r(\omega)] \\ &= W(\omega)[H_d(\omega) - Q(\omega)P(\omega)] \end{aligned}$$

$$= W(\omega)Q(\omega) \left[\frac{H_{dr}(\omega)}{Q(\omega)} - P(\omega) \right] \quad (8.2.67)$$

For mathematical convenience, we define a modified weighting function $\hat{W}(\omega)$ and a modified desired frequency response $\hat{H}_{dr}(\omega)$ as

$$\begin{aligned} \hat{W}(\omega) &= W(\omega)Q(\omega) \\ \hat{H}_{dr}(\omega) &= \frac{H_{dr}(\omega)}{Q(\omega)} \end{aligned} \quad (8.2.68)$$

Then the weighted approximation error may be expressed as

$$E(\omega) = \hat{W}(\omega)[\hat{H}_{dr}(\omega) - P(\omega)] \quad (8.2.69)$$

for all four different types of linear-phase FIR filters.

Given the error function $E(\omega)$, the Chebyshev approximation problem is basically to determine the filter parameters $\{\alpha(k)\}$ that minimize the maximum absolute value of $E(\omega)$ over the frequency bands in which the approximation is to be performed. In mathematical terms, we seek the solution to the problem

$$\min_{\text{over } \{\alpha(k)\}} \left[\max_{\omega \in S} |E(\omega)| \right] = \min_{\text{over } \{\alpha(k)\}} \left[\max_{\omega \in S} |\hat{W}(\omega)[\hat{H}_{dr}(\omega) - \sum_{k=0}^L \alpha(k) \cos \omega k]| \right] \quad (8.2.70)$$

where S represents the set (disjoint union) of frequency bands over which the optimization is to be performed. Basically, the set S consists of the passbands and stopbands of the desired filter.

The solution to this problem is due to Parks and McClellan (1972a), who applied a theorem in the theory of Chebyshev approximation. It is called the *alternation theorem*, which we state without proof.

Alternation Theorem: Let S be a compact subset of the interval $[0, \pi)$. A necessary and sufficient condition for

$$P(\omega) = \sum_{k=0}^L \alpha(k) \cos \omega k$$

to be the unique, best weighted Chebyshev approximation to $\hat{H}_{dr}(\omega)$ in S , is that the error function $E(\omega)$ exhibit at least $L + 2$ extremal frequencies in S . That is, there must exist at least $L + 2$ frequencies $\{\omega_i\}$ in S such that $\omega_1 < \omega_2 < \dots < \omega_{L+2}$, $E(\omega_i) = -E(\omega_{i+1})$, and

$$|E(\omega_i)| = \max_{\omega \in S} |E(\omega)| \quad i = 1, 2, \dots, L + 2$$

We note that the error function $E(\omega)$ alternates in sign between two successive extremal frequencies. Hence the theorem is called the alternation theorem.

To elaborate on the alternation theorem, let us consider the design of a lowpass filter with passband $0 \leq \omega \leq \omega_p$ and stopband $\omega_s \leq \omega \leq \pi$. Since the